When spatial statistics outperform machine learning: corridor-based ground-truth evaluation of crime risk prediction for pedestrian safety routing

Abstract

Kernel Density Estimation (KDE)—a spatial statistics technique developed in the 1960s—outperforms modern machine learning (ML) approaches including XGBoost and Graph Neural Networks (GNN) for pedestrian safety routing. This finding emerges from a novel evaluation framework that measures what existing assessments do not directly capture: actual crime avoidance on real routes. While prior work evaluates prediction methods using accuracy metrics on held-out crime counts, we introduce corridor-based evaluation that directly measures whether recommended 'safe' routes reduce pedestrian exposure to violent incidents compared to shortest-path alternatives. Applying this framework to 1.15 million street segments in Chicago with five prediction methods, strict temporal holdout across three test months, and 36,000 route evaluations per method, we find KDE achieves 54.5% greater crime avoidance than XGBoost and 53.3% greater than GNN. The performance hierarchy—KDE > GNN ≈ XGBoost > Getis-Ord Gi* > Historical Average—is consistent across all robustness checks. KDE's explicit encoding of spatial decay produces smooth, continuous risk surfaces ideally suited to routing optimization, which requires identifying gradients between alternative paths. ML models may produce discontinuous predictions that fail to capture these gradients. This work contributes a rigorous evaluation methodology and evidence that model complexity does not guarantee superior performance for spatially-structured downstream applications.
Keywords: pedestrian safety routing, crime risk prediction, kernel density estimation, spatial autocorrelation, ground truth evaluation

Key Policy Highlights

- Cities can deploy more effective pedestrian safety routing using Kernel Density Estimation, which requires no machine learning infrastructure—only crime incident coordinates and a spatial statistics library.
- Prediction accuracy on standard benchmarks does not translate to routing effectiveness; evaluation should measure actual crime avoidance on recommended routes.
- KDE-guided routes achieve 54.5% greater crime avoidance than XGBoost and 53.3% greater than Graph Neural Networks across 36,000 route evaluations in Chicago.
- The corridor-based ground truth evaluation framework introduced here is reproducible and applicable to any city with geocoded crime data and street network information.

1. Introduction

Kernel Density Estimation, a spatial statistics technique originating in the 1960s, substantially outperforms modern machine learning methods—including XGBoost and Graph Neural Networks—for pedestrian safety routing. This finding contradicts the prevailing assumption in crime prediction research that complex machine learning models necessarily improve upon traditional spatial statistical approaches. Recent studies consistently report that ML methods achieve higher accuracy than KDE on crime count prediction tasks, with one 2024 Chicago study finding Random Forest achieved higher performance than KDE. Yet when we evaluate these same methods on their actual purpose—generating routes that reduce pedestrian exposure to violent crime—the ranking appears to invert. Across 36,000 route evaluations per method on Chicago's complete walkable street network, KDE-guided routes average 0.578 corridor crimes avoided per route—54.5% more than XGBoost (0.374) and 53.3% more than GNN (0.377). The disconnect between prediction accuracy and routing effectiveness reveals a fundamental methodological gap: existing evaluations measure the wrong outcome. What matters for pedestrian safety routing is not how well a model predicts crime counts, but how well its risk surface captures the spatial gradients that distinguish safer paths from riskier alternatives.

The methodological gap we address is substantial. Existing evaluations measure how well models predict crime counts at specific locations, then assume—without verification—that superior prediction yields superior routing outcomes. Yet these are distinct objectives: crime count prediction optimizes accuracy at individual locations, while routing optimization minimizes total exposure across entire paths. A model may accurately predict segment-level counts yet fail to capture the spatial gradients critical for distinguishing safer routes from riskier alternatives. No prior work has evaluated crime prediction methods on what pedestrian safety systems ultimately require: actual reduction in crime exposure along recommended routes.

We address this gap by introducing corridor-based ground truth evaluation. For each origin-destination (OD) pair, we generate a shortest path and a safety-aware path incorporating predicted crime risk. We then construct spatial corridors around both routes and count violent crime incidents occurring within each corridor during a strictly held-out future period. The difference constitutes our ground truth measure of crime avoidance.

We implement this framework on Chicago's complete walkable street network (1.15 million street segments) with strict rolling temporal holdout across three test months, three corridor radii (15m, 30m, 50m), four time-of-day windows, and stratified origin-destination sampling. We compare five prediction methods: Historical Average, Hotspot Mapping, KDE, XGBoost, and Graph Neural Networks.

The empirical results establish a clear and consistent performance hierarchy: KDE > GNN ≈ XGBoost > Getis-Ord Gi* > Historical Average. KDE achieves a 36.0% win rate compared to 29.7% for GNN and 29.3% for XGBoost, with 53.3% greater mean crime avoidance than GNN and 54.5% greater than XGBoost. GNN and XGBoost are statistically indistinguishable ($p = 0.930$). All pairwise comparisons except GNN vs. XGBoost are statistically significant, and the hierarchy is stable across all robustness checks.

KDE's advantage derives from alignment between its assumptions and routing requirements: its Gaussian kernel produces smooth, continuous risk surfaces where adjacent streets receive gradually varying risk estimates—precisely what routing optimization requires to identify gradients between alternative paths.

This work makes three contributions. Methodologically, we introduce corridor-based ground truth evaluation—a rigorous, reproducible framework that directly measures crime avoidance on real routes rather than relying on prediction accuracy as a proxy. The framework is applicable to any city with geocoded crime data and street network information. Empirically, we provide the first systematic comparison of crime prediction methods evaluated on actual routing outcomes at scale: 36,000 route evaluations per method across three test months, three corridor radii, and four time-of-day windows. The finding that KDE substantially outperforms XGBoost and GNN challenges prevailing assumptions in both crime prediction research and applied machine learning. Theoretically, we demonstrate conditions under which explicit encoding of spatial structure outperforms learned representations: when spatial autocorrelation is well-characterized by theory, when downstream tasks optimize over spatial aggregates, when training data are sparse relative to output resolution, and when prediction and application objectives are misaligned.

These findings carry immediate practical value: cities seeking to deploy pedestrian safety routing can achieve superior performance using KDE, which requires no machine learning infrastructure, no model training, and no feature engineering—only crime incident coordinates and a standard spatial statistics library. More broadly, the disconnect between prediction accuracy and routing effectiveness reflects a fundamental misalignment between model training objectives and downstream application requirements.

2. Literature Review

Crime prediction has progressed from simple pin maps to sophisticated spatial modeling and machine learning, reflecting advances in computational capabilities and growing recognition that crime exhibits complex spatial and temporal patterns. The field has developed increasingly nuanced methods drawing on environmental criminology, spatial statistics, and geographic information science. Yet despite this methodological sophistication, a fundamental question remains underexplored: how well do crime prediction methods perform when applied to downstream spatial decision-making tasks such as pedestrian route planning?

2.1 Traditional Spatial Methods for Crime Prediction

Kernel Density Estimation has served as a foundational technique in crime analysis since Chainey, Tompson, and Uhlig (2008) systematically compared hotspot mapping techniques and established KDE's utility for predicting spatial crime patterns. The method places smooth kernels at each crime location, creating continuous risk surfaces that naturally encode spatial autocorrelation. Hu et al. (2018) extended this framework to incorporate temporal dimensions through spatio-temporal KDE with cross-validated bandwidth selection.

Hotspot mapping identifies discrete high-crime zones through spatial clustering. Weisburd's (2015) law of crime concentration demonstrated that crime clusters in remarkably small geographic areas, providing theoretical justification for hotspot-based strategies. However, hotspot mapping produces binary classifications, limiting its utility for applications requiring fine-grained risk distinctions. Risk Terrain Modeling (Caplan & Kennedy, 2011) offered an alternative integrating environmental criminology with spatial analysis. These traditional methods produce risk estimates through explicit statistical assumptions about spatial structure; the next section examines whether machine learning approaches improve upon them.

2.2 Machine Learning Approaches for Crime Prediction

Machine learning applications for crime prediction have grown substantially, driven by increasing urban data availability and algorithmic advances. Gradient boosting methods, particularly XGBoost and Random Forest, have emerged as popular choices due to their capacity to model complex, non-linear relationships among spatial, temporal, and contextual features (Kounadi et al., 2020). These approaches typically require extensive feature engineering, including distance-decay crime aggregations and temporal pattern encoding.

Graph Neural Networks represent the frontier of deep learning for crime prediction, treating street networks as graphs to enable information propagation across urban topology. Zhou et al. (2024) introduced HDM-GNN for capturing multi-scale spatial and temporal crime correlations, while Tekin and Kozat (2023) demonstrated graph-based models for high-resolution crime forecasting. The theoretical appeal lies in learning spatial relationships directly from network structure, potentially capturing patterns that elude hand-crafted features.

Wang et al. (2025) introduced MRAGNN for capturing multi-type crime correlations. These GNN methods report improvements on forecasting benchmarks, yet none assess performance on downstream routing applications—motivating our corridor-based evaluation framework.

Despite enthusiasm for machine learning, important caveats remain. Mandalapu et al. (2023) raised interpretability concerns with black-box models, and Safat, Asghar, and Gillani (2021) found that the best-performing algorithm varied across cities, suggesting no single approach consistently dominates. This echoes Makridakis et al. (2018), who demonstrated that traditional statistical methods often outperform complex machine learning in forecasting competitions. Yet whether superior prediction accuracy translates to superior downstream routing performance remains untested.

2.3 Pedestrian Safety Routing Systems

A separate literature has emerged around safety-aware pedestrian routing, connecting the crime prediction methods reviewed in Sections 2.1–2.2 to practical navigation applications. These systems consume crime risk estimates as inputs to route optimization, yet each adopts a single prediction method without comparative evaluation—treating the choice of prediction model as incidental rather than consequential for routing outcomes. Sohrabi et al. (2022) synthesized

research on safety in route-finding across multiple domains, identifying critical gaps in integrating safety into navigation systems and noting that most commercial applications optimize solely for travel time and distance.

Levy et al. (2020) developed SafeRoute, using deep reinforcement learning to balance safety and efficiency, improving distance from crimes by up to 17%—though evaluation focused on distance from incidents rather than actual crime exposure. Basu et al. (2023) found that safety perceptions significantly influence pedestrian route choice, with effects varying between day and night walking.

International research has extended safe routing to diverse contexts: Theron and Breetzke (2024) demonstrated that route recommendations differ substantially depending on crime data incorporation, while Berhanu et al. (2024) applied prediction-then-routing frameworks to traffic accident avoidance. Notably, none of these systems systematically compare prediction methods—each adopts a single approach without evaluating whether alternatives would yield safer routes.

2.4 The Evaluation Methodology Gap

A fundamental disconnect exists between how crime prediction methods are assessed and how they are ultimately used. Dominant evaluation metrics—Predictive Accuracy Index (Chainey et al., 2008), hit rate, RMSE, MAE, and AUC—all assess how well a model predicts crime counts at specific locations (Drawve & Wooditch, 2019; Kounadi et al., 2020).

However, these metrics measure prediction accuracy, not downstream application effectiveness. For pedestrian safety routing, what matters is not how well a model predicts crime counts, but whether its risk surface captures the spatial gradients distinguishing safer from riskier paths. A model achieving high PAI may fail to capture these gradients—information invisible to point-prediction metrics. This disconnect is exacerbated by crime data sparsity: most segments experience zero incidents monthly, dominating accuracy metrics with trivial zero predictions. Elmachtoub and Grigas (2022) formalized this as the ‘predict, then optimize’ problem, demonstrating that minimizing prediction error does not minimize downstream decision loss. Pedestrian safety routing is precisely such a setting: predictions are consumed by a shortest-path optimizer, and the relevant loss is crime exposure along recommended routes. This motivates an evaluation framework that directly measures routing outcomes.

2.5 Spatial Autocorrelation and the Case for Explicit Encoding

Understanding why certain methods may better serve routing applications requires examining spatial autocorrelation—the tendency for nearby locations to exhibit similar crime risk—which has direct implications for prediction method design. Mohler et al. (2011) demonstrated that crime exhibits clustering patterns analogous to earthquake aftershocks, with incidents at one location elevating risk at nearby locations through self-exciting point process dynamics. This

theoretical insight implies that effective crime risk estimation must account for the spatial spillover of risk, not merely predict counts at isolated locations.

KDE explicitly encodes spatial decay through its kernel function, ensuring that risk surfaces are inherently smooth and that nearby locations receive correlated risk estimates. Machine learning models, by contrast, must learn spatial relationships from data through feature engineering or architectural design choices. Wheeler and Steenbeek (2021) demonstrated that Random Forests could outperform traditional methods for long-term crime forecasting when carefully implemented, but emphasized the importance of using interpretability tools like SHAP to understand learned spatial patterns. Learned spatial representations may not capture the smooth gradients distinguishing safer from riskier routing alternatives: tree-based models produce discontinuous predictions at decision boundaries, while graph neural networks may struggle to learn specific distance-decay functions. Whether this theoretical advantage of explicit spatial encoding translates to measurable differences in routing outcomes is an empirical question that motivates the comparative evaluation developed in this paper.

2.6 Research Gap and Contributions

While substantial research has advanced crime prediction methods and separate work has developed safety-aware routing systems, a critical methodological gap persists: no systematic framework exists for evaluating whether improved crime prediction translates to actual crime avoidance in routing applications. Crime prediction evaluations rely on accuracy metrics divorced from routing outcomes; safety routing papers propose systems but do not rigorously compare prediction methods; and claims of machine learning superiority rest on evidence from prediction tasks rather than downstream applications. This paper addresses these gaps by introducing a corridor-based ground truth evaluation framework, conducting the first rigorous comparison of prediction methods evaluated on routing outcomes at scale, and providing theoretical insight into why explicit spatial encoding may outperform learned representations for this specific application domain.

3. Methods

3.1 Study Area and Data

This study uses the complete walkable street network of Chicago, comprising 1.15 million edge segments derived from OpenStreetMap (OpenStreetMap contributors, 2024). Long edges were split at uniform 50-meter intervals to ensure consistent segment lengths for risk attribution—without this step, longer streets would accumulate disproportionately high crime counts simply due to their greater spatial extent. Crime data includes over 750,000 geocoded incidents from the Chicago Data Portal. We retain violent crime types (homicide, assault, robbery, and battery) occurring in street-relevant locations, filtering by the portal's location description field to include only incidents on streets, sidewalks, alleys, and other public outdoor spaces; this excludes indoor

crimes such as domestic battery from the analysis. The resulting incidents are attributed to street segments using nearest-edge assignment with a 100-meter maximum snap distance.

The temporal scope uses a strict rolling holdout design: for each test month, training data includes all incidents from January 2023 through the month immediately preceding the test period. Specifically, models tested on July 2025 were trained on data through June 2025; models tested on October 2025 were trained through September 2025; and models tested on November 2025 were trained through October 2025. This temporal separation ensures that evaluation measures true predictive performance rather than in-sample fit, while the multi-month testing design assesses generalization across seasonal variations. The rolling design also mirrors real-world deployment conditions, where any operational system would necessarily predict future risk using only past observations.

3.2 Crime Risk Prediction Methods

We compare five risk prediction methods spanning the methodological spectrum from simple baselines to contemporary deep learning approaches. Each method produces edge-level risk scores that are transformed into routing costs via the following formula:

$$cost(e) = length(e) + \alpha \cdot \log(1 + \hat{r}(e)) \quad (1)$$

where α controls the tradeoff between route efficiency and safety. The logarithmic transformation prevents excessive sensitivity to extreme risk values while preserving rank ordering.

3.2.1 Historical Average

The Historical Average baseline represents the simplest approach to crime risk prediction. For each street segment in the network, we compute the mean number of violent crimes occurring on that specific edge during the training period, normalized by the training period duration to produce a crime rate per day. This approach fails to account for spatial autocorrelation and suffers from data sparsity, as most edges experience zero violent crimes during any given training period.

where V_{train} is the set of training incidents, edge(c) denotes the nearest edge to crime c, and T is the number of days in the training period. This creates independent risk scores where each edge's prediction depends solely on its own historical crime count, with no information borrowed from neighboring streets.

3.2.2 Hotspot Mapping (Getis-Ord Gi*)

The Hotspot baseline identifies statistically significant clusters of high-crime edges using the Getis-Ord Gi* statistic (Getis & Ord, 1992; Ord & Getis, 1995), a standard method in spatial crime analysis. For each edge in the network, Gi* compares the local sum of crime counts within a distance band to the global mean, producing a z-score that indicates whether the local

concentration is significantly higher (hotspot) or lower (coldspot) than expected under spatial randomness:

$$G_i^* = \frac{\sum_j w_{ij} x_j - \bar{x} \sum_j w_{ij}}{s \sqrt{\frac{n \sum_j w_{ij} - \left(\sum_j w_{ij}\right)^2}{n-1}}}$$

where xj is the crime count on edge j, wij is a binary spatial weight equal to 1 if edges i and j are within 300 meters (0 otherwise), and $\bar{X}$ and S are the global mean and standard deviation of crime counts across all edges.

We compute Gi* using binary distance-band weights with a 300-meter threshold applied to edge centroids projected in UTM Zone 16N (EPSG:32616). Each edge's crime count is derived from the same incident data used by other methods. The resulting z-scores provide a continuous measure of local spatial clustering intensity, with higher values indicating stronger hotspot membership.

To integrate Gi* into the routing framework, we shift z-scores to a non-negative range and scale them to match the Historical Average baseline mean, ensuring comparability across methods under the same α. This continuous surface replaces the binary hotspot indicator used in preliminary analyses.

3.2.3 Kernel Density Estimation (KDE)

KDE is a nonparametric spatial statistics method (Silverman, 1986) that estimates a continuous probability density surface from point data. The method places a Gaussian kernel—a smooth, bell-shaped probability distribution—centered at each crime location. Risk at any point $x \in \mathbb{R}^2$ is computed as the sum of contributions from all crime incidents:

$$f(x) = \frac{1}{nh^2} \sum_{c \in V_{\text{train}}} K\left(\frac{\|x - \text{loc}(c)\|}{h}\right) \tag{2}$$

where $n = | V_{train} |$ is the number of training incidents, h is the bandwidth parameter controlling smoothing, and $K$is the Gaussian kernel function:

$$K(u) = \frac{1}{\sqrt{2\pi}} exp\left(-\frac{u^2}{2}\right)$$

The bandwidth parameter is critical: too small and the surface becomes spiky with isolated peaks at each crime location; too large and meaningful local variation is smoothed away. We use a bandwidth of $h = 500$ meters, consistent with established practice in crime analysis where bandwidths of 400–600 meters are commonly employed for neighborhood-level pattern detection (Chainey et al., 2008; Hu et al., 2018).

To assign risk to street edges, we evaluate the KDE surface at each edge centroid:

$$\hat{r}_{KDE}(e) = f(\hat{x}_e) \tag{3}$$

where $\hat{x}_e$ denotes the geometric centroid of edge e. This approach naturally handles data sparsity through spatial smoothing—even streets with no observed crimes receive non-zero risk estimates based on nearby incidents—and explicitly encodes the principle that risk decays smoothly with distance. For comparability across methods, KDE risk scores are rescaled so that their mean matches the mean of the Historical Average baseline.

To assess bandwidth sensitivity, we also evaluate KDE at 250m and 1000m bandwidths; results are reported in Section 4.4.

3.2.4 XGBoost with Distance-Weighted Features

The XGBoost model (Chen & Guestrin, 2016) uses gradient-boosted decision trees trained on engineered features that capture spatial, temporal, and network characteristics of each street edge.

Distance-Weighted Crime Attribution: The key innovation is distance-weighted risk aggregation using exponential decay. For each crime incident c with location $\mathrm{loc}(c)$, we attribute it to all edges within a maximum radius $R_{max} = 50$ meters. Let $d(c, e)$ denote the Euclidean distance from the crime location to the nearest point on edge e:

$$d(c, e) = \min_{y \in \gamma(e)} \|\mathrm{loc}(c) - y\|_2$$

The attribution weight follows an exponential decay function:

$$w(c, e) = \begin{cases} exp\left(-\dfrac{d(c, e)}{\lambda}\right) & \text{if } d(c, e) \leq R_{\max} \\ 0 & \text{otherwise} \end{cases}$$

where $\lambda = R_{max}/3 \approx 16.67$ meters is the decay constant. This yields weights of 1.0 at distance 0, approximately 0.5 at 11.5 meters, and approximately 0.05 at the 50-meter boundary. The total weighted crime count for edge e is:

$$\tilde{r}(e) = \sum_{c \in V_{\text{train}}} w(c, e)$$

Note that weights are not normalized per crime—a single incident can contribute total weight greater than 1 across all nearby edges. This design reflects the assumption that crime risk diffuses spatially, elevating risk on all nearby streets rather than being confined to the single nearest edge, consistent with crime pattern theory (Brantingham & Brantingham, 1993).

This attribution creates a structural asymmetry: KDE distributes each crime's probability mass via a unit-integral Gaussian kernel, whereas the distance-weighted scheme can assign total weight greater than 1 per crime. The Pareto frontier evaluation (Section 4.5) resolves scale differences by comparing methods at matched detour levels, and a per-crime normalization robustness check (Section 5.1) confirms that constraining ML weights to sum to 1 degrades performance.

Model Specification: Additional features include temporal patterns (month, day of week, 6-hour time bin), edge characteristics (length, road type), and lagged crime indicators. The model uses Poisson regression loss with $\log(l(e))$ as an offset term, ensuring predictions scale appropriately with segment length. Training uses validation with early stopping (patience of 500 rounds) to prevent overfitting.

3.2.5 Graph Neural Network (GNN)

The GNN architecture combines distance-weighted input features (identical to XGBoost, using the same exponential decay attribution described in Section 3.2.4) with learned spatial smoothing through graph convolution, enabling the model to aggregate information across the street network topology.

Dual Graph Representation: We represent the street network as a dual graph $G' = (V', E')$, where each street edge $e \in E$ becomes a node $v_e \in V'$, and two nodes are connected if their corresponding edges share an intersection in the original network:

$$E' = \{(v_e, v_{e'}) : e \text{ and } e' \text{ share a common endpoint}\}$$

This representation allows the GNN to propagate information along the network structure—risk on one street can influence predictions for topologically connected streets.

Architecture: We evaluated three GNN architectures—GraphSAGE (Hamilton et al., 2017), GATv2 dynamic attention convolution (Brody et al., 2022), and multi-hop diffusion convolution combining transition-matrix propagation (Atwood & Towsley, 2016) with learned per-hop mixing (Abu-El-Haija et al., 2019)—and selected GATv2 based on superior validation performance. GATv2 corrects a limitation of the original GAT (Veličković et al., 2018) whose attention scores are static—the ranking of neighbor importance is identical for every query node. GATv2 applies the learned weight matrix before the nonlinearity, yielding dynamic attention where each node can attend differently to its neighbors:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(a^T[Wh_i \parallel Wh_j]\right)\right)}{\sum_{k \in N(i)} \exp\left(\text{LeakyReLU}\left(a^T[Wh_i \parallel Wh_k]\right)\right)}$$

where h_i and h_j are node representations, W is a learned weight matrix, a is a learned attention vector, ‖ denotes concatenation, and α_{ij} are softmax-normalized attention coefficients. The model uses 4 parallel attention heads with d = 32 dimensions each (128 total), concatenated to preserve the hidden dimension. Four GATv2 layers are stacked with residual connections, layer

normalization, ReLU activation, and dropout (p = 0.2) between each layer. Input features are projected to the hidden dimension via a linear layer, and the final representation is passed through a linear projection to scalar log-rate predictions.

Training: Mini-batch training with neighbor sampling (Hamilton et al., 2017) enables efficient training on the 1.15 million-node dual graph. The model uses Poisson negative log-likelihood loss with Adam optimization, gradient clipping, and early stopping with validation on held-out time-slice graphs. Raw predictions are calibrated using isotonic regression on a held-out calibration set, ensuring predicted risk values align with observed crime frequencies.

3.3 Evaluation Framework

We introduce corridor-based ground truth evaluation that directly measures crime avoidance rather than prediction accuracy. This framework addresses a fundamental limitation of existing evaluations: prediction metrics (RMSE, AUC, PAI) measure how well models estimate crime counts on individual edges, but routing effectiveness depends on how well risk surfaces capture gradients between alternative paths.

3.3.1 Route Generation

Let $G = (V, E)$ denote the street network graph with vertex set V(intersections) and edge set E(street segments). For each edge $e \in E$, let $l(e)$ denote its length in meters. Let $\mathcal{P}(o, d)$ denote the set of all simple paths from origin o to destination d, where each path $P \in \mathcal{P}(\mathrm{o}, \mathrm{d})$ is an ordered sequence of edges, such that $e \in P$.

For each origin-destination pair (o, d), we generate two routes using Dijkstra's algorithm (Dijkstra, 1959):

Shortest path: The distance-minimizing route serves as the baseline, representing a pedestrian navigating without safety information:

$$P_{\text{short}} = \underset{P \in \mathcal{P}(o,d)}{\arg\min} \sum_{e \in P} l(e)$$

Safety-aware path: The risk-weighted route incorporates predicted crime risk from method M with risk aversion parameter α:

$$P_{\text{safe}}^{M} = \underset{P \in \mathcal{P}(o,d)}{\arg\min} \sum_{e \in P} \left[l(e) + \alpha \cdot \log\left(1 + \hat{r}_M(e)\right) \right]$$

where $\hat{r}_M(e)$ is method M's predicted crime risk for edge e. The logarithmic transformation ensures diminishing marginal penalty for increasingly risky edges, preventing excessive detours around extreme predictions while preserving rank ordering of risk.

3.3.2 Corridor Construction

For a route $P = (e_1, e_2, \dots, e_n)$ represented as an ordered sequence of edges, let $\gamma(e_i) \subset \mathbb{R}^2$ denote the geometric embedding of edge e_i as a line segment in projected coordinate space. All spatial operations use Universal Transverse Mercator Zone 16N (EPSG:32616), which provides sub-meter accuracy across the study area. The route geometry is the union of all edge geometries:

$$\gamma(P) = \bigcup_{i=1}^{n} \gamma(e_i)$$

We define the spatial corridor $C(P, R)$ as the Minkowski sum of the route geometry with a closed disk of radius R:

$$C(P, R) = \gamma(P) \oplus B_R(0)$$

where $B_R = \{x \in \mathbb{R}^2 : \| x \| \leq R\}$ is the closed disk of radius R centered at the origin and $\oplus$ denotes the Minkowski sum. Equivalently:

$$C(P, R) = \left\{x \in \mathbb{R}^2 : \ d\big(x, \gamma(P)\big) \leq R\right\}$$

where $d(x, \gamma(P)) = \min_{y \in \gamma(P)} \| x - y \|_2$ is the Euclidean distance from point x to the nearest point on the route geometry. This corridor represents the exposure zone—the spatial region within which incidents pose potential risk to a pedestrian traversing route P.

3.3.3 Crime Attribution

Let $V(t, \tau)$ denote the set of violent crime incidents occurring during evaluation period t (e.g., July 2025) within time-of-day window τ (e.g., 00:00–06:00). Each incident $v \in V(t, \tau)$ has geocoded coordinates $\mathrm{loc}(v) \in \mathbb{R}^2$. We define the corridor crime count as:

$$N(P, R, t, \tau) = \sum_{v \in V(t,\tau)} \mathbb{1}[\mathrm{loc}(v) \in C(P, R)]$$

where $\mathbb{1}[\cdot]$ is the indicator function. This counts incidents whose locations fall within the corridor polygon during the specified evaluation window. Crime avoidance is thus computed over all incidents occurring within the corridor during the full evaluation month and time-of-day window, representing cumulative monthly exposure reduction rather than single-trip risk. A pedestrian walking the route on a single day would expect exposure proportional to this value divided by the number of days in the evaluation period.

3.3.4 Crime Avoidance Metric

The ground truth measure of routing effectiveness is crime avoidance—the reduction in corridor crime exposure achieved by the safety-aware route relative to the shortest-path baseline:

$$A_M(o, d, R, t, \tau) = N(P_{\text{short}}, R, t, \tau) - N\left(P_{\text{safe}}^M, R, t, \tau\right)$$

Interpretation:

- $A_M > 0$: The safety-aware route reduced crime exposure (success)
- $A_M < 0$: The safety-aware route increased crime exposure (failure)
- $A_M = 0$: Equivalent exposure

3.3.5 Aggregate Performance Metrics

We aggregate across K origin-destination pairs to obtain summary statistics for each prediction method:

Mean crime avoidance measures the average monthly reduction in corridor crimes per evaluated route:

$$\hat{A}_M = \frac{1}{K}\sum_{k=1}^{K} A_M(o_k, d_k, R, t, \tau)$$

Win rate measures the proportion of routes achieving strictly positive crime avoidance:

$$W_M = \frac{1}{K}\sum_{k=1}^{K} \mathbb{1}[A_M(o_k, d_k, R, t, \tau) > 0]$$

Total crime avoidance measures aggregate crime reduction across all evaluated routes. Because route corridors overlap spatially and the sum scales linearly with the number of evaluations, this quantity is not interpretable as a count of distinct crimes avoided citywide; it serves as an intermediate quantity from which per-route means and percentage comparisons are derived:

$$T_M = \sum_{k=1}^{K} A_M(o_k, d_k, R, t, \tau)$$

3.4 Experimental Design

The experimental design incorporates multiple robustness checks to ensure findings generalize across evaluation parameters.

Origin-Destination Sampling: We evaluate 1,000 origin-destination pairs using stratified sampling: 500 pairs with origins and destinations sampled from six defined high-crime zones (Loop, Magnificent Mile, River North, Wicker Park, Englewood, and Austin), and 500 pairs sampled uniformly at random from all network nodes. All pairs are filtered to ensure minimum network distance of 1,000 meters and maximum of 5,000 meters, representing realistic pedestrian trip lengths.

Corridor Radii: We evaluate three corridor radii—$R \in \{15,30,50\}$ meters—to assess sensitivity to spatial attribution assumptions. Consistent findings across radii indicate robustness to this methodological choice.

Time-of-Day Segmentation: Routes are evaluated separately for four 6-hour windows: Overnight (00:00–05:59), Morning (06:00–11:59), Afternoon (12:00–17:59), and Evening (18:00–23:59), capturing variation in crime patterns and pedestrian vulnerability across the full day.

Route Distance Robustness: A preliminary evaluation using unconstrained origin-destination pairs (mean network distance approximately 18 km) confirmed that the performance hierarchy is preserved across an order-of-magnitude range in route distances, with KDE's relative advantage over XGBoost remarkably stable.

Fair Comparison via Pareto Frontier Analysis: Different prediction methods produce risk scores on vastly different scales. We address this through Pareto frontier analysis: for each method, we sweep α across a logarithmic range and compute the resulting (detour percentage, crime avoidance) pairs. Comparing methods at matched detour levels (e.g., 5% average detour) ensures fair assessment independent of risk score scaling.

Statistical Testing: Differences between methods are assessed using Mann-Whitney U tests with Bonferroni correction. All methods are evaluated on identical OD pairs, corridors, time periods, and test months to ensure strict comparability.

4. Results

4.1 Overall Performance

Table 1 presents headline performance metrics for all five prediction methods across 36,000 route evaluations per method—1,000 OD pairs evaluated under 36 conditions (3 months × 3 corridor radii × 4 time windows). Observations within OD pairs are not independent, as the same origin-destination geometry is evaluated under varying conditions (Table A.2).

Table 1: Overall Performance by Prediction Method

Method	Mean Avoided	Win Rate (%)	Mean Detour (%)	KDE Adv. (%)
KDE	0.578	36.0	11.58	—
GNN	0.377	29.7	10.25	53.3

XGBoost	0.374	29.3	8.34	54.5
Getis-Ord Gi*	0.329	27.3	9.32	75.6
Historical Avg	0.045	11.8	1.64	1185.8

Note: N = 36,000 route evaluations per method. Win Rate = percentage of routes where the safe route encounters fewer corridor crimes than the shortest path. KDE Advantage = percentage improvement in mean crimes avoided relative to each alternative method.

KDE-guided routes averaged 0.578 crimes avoided per route—54.5% more than XGBoost (0.374) and 53.3% more than GNN (0.377). All methods show a median of 0.0—reflecting crime sparsity along 1–5 km routes—but KDE achieves the highest mean (0.578) compared to 0.377 for GNN and 0.374 for XGBoost (Figure 1). KDE win rate of 36.0% (Figure 2) leads all methods. No method exceeds 50%, reflecting the high tie rate inherent to short routes with sparse crime. KDE is the only method to significantly exceed 30% win rate (binomial test, $p < 0.001$; Table A.5), while Historical Average achieves only 11.8%. KDE also exhibits the lowest loss rate at 7.3% (Figure A.4).

Figure 1. Mean crimes avoided per route by prediction method. Error bars show 95% bootstrap confidence intervals. N = 36,000 routes per method.

Figure 2. Win rate (percentage of routes where the safe route encounters fewer corridor crimes than the shortest path) by prediction method.

4.2 Statistical Comparisons

Mann-Whitney U tests confirm the statistical significance of performance differences (Table 2). All pairwise comparisons except GNN vs. XGBoost are statistically significant and survive Bonferroni correction (alpha = 0.005). KDE vs. XGBoost ($p = 7.48e-141$), KDE vs. GNN ($p = 1.09e-137$), and KDE vs. Getis-Ord Gi* ($p = 5.61e-210$) are all significant at $p < 0.001$. GNN versus XGBoost is not statistically significant ($p = 0.930$), indicating these methods are effectively indistinguishable in routing performance. The 95% bootstrap confidence intervals (see also Figure A.3 for empirical cumulative distribution functions confirming KDE's stochastic dominance) (Figure A.2, Table A.3) reinforce this finding: KDE confidence interval [0.564, 0.591] does not overlap with any other method, while GNN [0.364, 0.390] and XGBoost [0.361, 0.387] intervals overlap substantially.

Table 2: Pairwise Statistical Comparisons (Mann–Whitney U Tests)

Comparison	U	p-value	r	Sig
KDE vs GNN	710,577,099	1.09e-137	-0.097	***†
KDE vs XGBoost	711,135,177	7.48e-141	-0.097	***†
KDE vs Getis-Ord Gi*	724,950,534	5.61e-210	-0.119	***†
KDE vs Historical Avg	804,440,454	< 1e-300	-0.241	***†

GNN vs XGBoost	648,219,858	0.930	-0.000	ns
GNN vs Getis-Ord Gi*	661,892,477	2.25e-08	-0.021	***†
GNN vs Historical Avg	734,120,920	< 1e-300	-0.133	***†
XGBoost vs Getis-Ord Gi*	661,720,265	3.05e-08	-0.021	***†
XGBoost vs Historical Avg	734,271,460	< 1e-300	-0.133	***†
Getis-Ord Gi* vs Historical Avg	719,464,559	1.76e-217	-0.110	***†

*Note: Non-parametric Mann–Whitney U comparisons. All tests survive Bonferroni correction (α = 0.005 for 10 comparisons). r = rank-biserial correlation (effect size). *** p < .001; ** p < .01; † survives Bonferroni correction.*

4.3 Effect Sizes

Cohen's d effect sizes quantify the practical significance of each method's crime avoidance relative to zero—the performance of naive shortest-path routing with no safety optimization—rather than between methods, as pairwise differences are already captured by the Mann-Whitney U tests above (Table A.6). KDE achieves a small effect size (d = 0.45), the largest among all methods. GNN (d = 0.30) and XGBoost (d = 0.30) show nearly identical small effect sizes, as does Getis-Ord Gi* (d = 0.26). Historical Average shows a negligible effect size (d = 0.05), consistent with its failure to systematically outperform shortest-path routing. The effect size hierarchy mirrors the crime avoidance ranking, confirming that performance differences are practically meaningful as well as statistically significant.

4.4 Robustness Checks

We assess the robustness of findings across multiple experimental conditions.

Corridor Radii: Performance at corridor radii of 15m, 30m, and 50m shows consistent method rankings (Figure A.1). KDE ranks first at all three radii, with absolute crime avoidance increasing with corridor width as expected (wider corridors capture more crime events). The relative performance hierarchy remains invariant to this evaluation parameter.

KDE Bandwidth Sensitivity: To verify that KDE's advantage is not an artifact of a particular bandwidth choice, we evaluate three bandwidths (250m, 500m, 1000m) through the full routing evaluation pipeline. Performance is stable across the four-fold bandwidth range: the 250m bandwidth produces approximately 1% less crime avoidance than the primary 500m specification, while the 1000m bandwidth produces approximately 4% less. All three bandwidths substantially outperform XGBoost and GNN, confirming that KDE's advantage is robust to bandwidth choice.

Temporal Consistency: Monthly breakdown across July, October, and November 2025 confirms KDE ranks first in all three test months (Table A.4, Figure A.5). July shows the highest crime avoidance for most methods, likely reflecting seasonal patterns, while the performance hierarchy

remains consistent. Crime avoidance varies across time-of-day bins, with evening and overnight periods showing higher avoidance than daytime hours across all methods, reflecting higher crime rates during those periods.

OD Pair Types: Comparison between hotspot-targeted and random origin-destination pairs (Figure A.6) reveals that hotspot OD pairs show higher absolute crime avoidance than random OD pairs (KDE: 0.764 vs 0.391 mean crimes avoided). With pedestrian-realistic 1–5 km routes, hotspot OD pairs traverse crime-dense areas where safety-aware routing can find meaningfully different paths, while random OD pairs often traverse low-crime areas where both paths encounter zero incidents. KDE maintains top ranking for both OD pair types.

Ranking Consistency: The full performance hierarchy is consistent across all twelve experimental conditions (4 time bins × 3 corridor radii; Table A.4, Figure A.1), with GNN and XGBoost consistently occupying second and third positions (statistically indistinguishable from each other).

Crime Type Filtering: Training on all violent crimes—including indoor incidents such as domestic battery—while evaluating on street crimes only did not improve performance for any method. This confirms that the street-relevant location filter (Section 3.1) removes noise rather than discarding useful signal for street-level risk estimation.

4.5 Pareto Frontier Analysis

The Pareto frontier (Figure 3, Table A.8) reveals safety-detour tradeoffs across the full range of risk aversion parameters. KDE dominates the Pareto frontier at all operating points across all time-of-day bins. At approximately 5% mean detour, KDE achieves substantially higher crime avoidance than XGBoost and GNN across all time bins. At approximately 10% mean detour, KDE maintains this advantage. Diminishing returns are evident beyond 10% detour, as curves flatten at high alpha values—additional detour yields progressively smaller safety gains. Evening and overnight bins show higher crime avoidance than daytime bins at equivalent detour levels, reflecting higher baseline crime rates during those periods. Alpha parameter sensitivity analysis (Figure A.7) confirms that optimal alpha values differ substantially across methods due to differences in risk score magnitudes, validating our approach of method-specific alpha tuning for fair comparison. Example route comparisons illustrate these patterns: in Englewood, a KDE-guided route achieves 100% crime reduction with only 5 meters additional distance (Figure A.8).

Figure 3. Pareto frontier showing safety–detour tradeoffs across risk aversion parameters across four time-of-day windows (6-hour bins).

4.6 Spatial Performance Analysis

To assess whether routing performance varies systematically across neighborhoods, we analyze the spatial distribution of crime avoidance. Performance residuals exhibit significant positive spatial autocorrelation for all methods except Historical Average (Table 3), indicating that crime

avoidance clusters in high-crime neighborhoods where routing differentiation is most feasible. This spatial clustering reflects variation in crime density: areas with higher crime rates (West Side, South Side) offer more opportunity for safety-aware routing to reduce exposure, representing an expected property of the evaluation framework rather than evidence of model deficiency. KDE shows the highest spatial clustering (Moran's I = 0.291), followed by GNN (0.277) and XGBoost (0.249), while Historical Average exhibits no significant spatial structure (0.143, p = 0.017). The consistent clustering across prediction-based methods suggests that crime avoidance is spatially concentrated regardless of the underlying model, with areas of higher crime density offering more opportunity for routing differentiation (Figure A.9, Table A.9).

While crime density variation explains much of this spatial clustering, unmeasured neighborhood-level factors likely contribute to spatial heterogeneity in model performance. Land use composition, street lighting density, pedestrian traffic volume, and the built environment's permeability to through-movement all shape both crime occurrence and the feasibility of alternative routing, yet none are explicitly modeled here. Areas where these latent factors diverge from crime density patterns—such as well-lit commercial corridors with high crime counts but low actual pedestrian risk—may produce spatially structured prediction errors. Future work incorporating such covariates could reduce the residual spatial autocorrelation observed across all methods and improve routing recommendations in neighborhoods where current models systematically over- or under-estimate walkable risk.

Table 3: Spatial Autocorrelation of Crime Avoidance Residuals (Moran's I)

Method	Moran's I	Expected I	z-score	p-value	n
KDE	0.2913	-0.0137	4.1446	0.001	74
GNN	0.2771	-0.0137	4.0593	0.001	74
XGBoost	0.2490	-0.0137	3.5932	0.001	74
Getis-Ord Gi*	0.2415	-0.0137	3.6956	0.001	74
Historical Avg	0.1430	-0.0137	2.1971	0.017	74

Note: Queen contiguity weights. Expected I under spatial randomness = −0.014 (n = 74 community areas). Significance assessed via 999 random permutations.

Figure A.9. Mean crime avoidance by community area for each method, with KDE exhibiting the broadest geographic coverage of high avoidance. KDE achieves the highest mean crime avoidance in the majority of areas.

5. Discussion

5.1 Why Does KDE Outperform Machine Learning?

The central finding of this study—that Kernel Density Estimation substantially outperforms both gradient-boosted trees and graph neural networks for pedestrian safety routing—demands

theoretical explanation. We propose that KDE's advantage derives from three interrelated properties: continuous risk surfaces, explicit spatial decay encoding, and alignment between method assumptions and application requirements.

First, KDE produces inherently continuous risk surfaces where adjacent street segments receive smoothly varying risk estimates. This continuity matters critically for routing optimization, which must identify gradients between alternative paths. When comparing two routes that diverge for several blocks before reconverging, the routing algorithm needs risk estimates that reflect the cumulative difference in exposure across all traversed segments. KDE's smooth surfaces ensure that small spatial displacements produce proportionally small risk differences, enabling the router to make fine-grained distinctions between nearby alternatives. Tree-based models like XGBoost, by contrast, produce discontinuous predictions at decision boundaries. Two adjacent street segments may fall on opposite sides of a split in the decision tree, receiving markedly different risk estimates despite their spatial proximity. These discontinuities can cause routing algorithms to make suboptimal choices, either overweighting small risk differences that reflect decision boundary artifacts or failing to distinguish genuinely different risk levels when both segments fall in the same leaf node.

Second, KDE explicitly encodes the principle that crime risk decays smoothly with distance from incident locations—a well-established empirical regularity in environmental criminology. The Gaussian kernel function embeds this spatial decay directly into the method's mathematical structure, requiring no learning from data. Machine learning models must instead discover spatial relationships through training, either via hand-crafted distance-decay features (as in our XGBoost implementation) or through learned representations (as in the GNN). This learning process introduces two potential failure modes: the model may learn incorrect spatial decay functions that do not match the true crime risk distribution, or it may fail to generalize learned patterns to areas with sparse training data. KDE sidesteps both problems by assuming a theoretically-motivated decay function that applies uniformly across the study area.

Third, and perhaps most importantly, KDE's assumptions align precisely with the requirements of routing optimization in ways that machine learning objectives do not. KDE optimizes for smooth interpolation across space—exactly what routing requires. XGBoost optimizes for minimizing prediction error on held-out crime counts, which may produce accurate point estimates while failing to capture the spatial gradients that matter for routing. The GNN optimizes for node-level prediction accuracy on the street network graph, which again prioritizes individual segment accuracy over the relative comparisons that routing demands. This misalignment between training objective and downstream application explains why models achieving competitive accuracy on prediction benchmarks nonetheless underperform KDE on routing tasks.

XGBoost was given every reasonable advantage: extensive hyperparameter optimization across 30+ configurations, 166 engineered features spanning multiple temporal windows, and multiple

post-processing approaches including Laplacian smoothing and isotonic calibration. That an untuned classical method outperforms this extensively optimized pipeline strengthens the central finding. Furthermore, we tested per-crime normalization of the distance-weighted attribution (constraining each crime's total attribution to sum to 1.0), which reduced XGBoost mean crime avoidance by 12%. This confirms that the unnormalized design captures genuine spatial risk diffusion; normalizing away this property degrades performance rather than removing an unfair advantage. Yet XGBoost's tree-based architecture produces fundamentally discontinuous risk surfaces—piecewise-constant approximations that cannot represent smooth spatial gradients as faithfully as KDE's continuous kernel surface.

This reflects the 'predict, then optimize' problem (Elmachtoub & Grigas, 2022): minimizing prediction error fails when accuracy is misaligned with the downstream objective. KDE succeeds not because it predicts crime counts more accurately—it likely does not—but because its smooth output produces risk gradients aligned with routing optimization. KDE is decision-optimal while ML methods are merely prediction-oriented.

5.2 The GNN Paradox: Why Network Structure Did Not Help

The GNN's underperformance warrants discussion, as GNNs would seem architecturally well-suited to graph-structured street networks with spatially-dependent crime. Yet GNN performs indistinguishably from XGBoost and substantially worse than KDE.

We identify four potential explanations. First, crime risk depends primarily on proximity to historical incidents—a fundamentally local property well-captured by distance-based features. Graph structure may add noise when the spatial process operates in Euclidean space rather than along network paths. Second, the 1.15 million-node dual graph creates a challenging optimization landscape. Third, GNNs are susceptible to oversmoothing (Rusch et al., 2023), where repeated message passing causes node representations to converge.

Fourth, despite 128 hidden channels, 4 attention heads, and 4 stacked layers, the GATv2 model could not match KDE. Further architecture search might improve GNN results, but diminishing returns relative to KDE's simplicity underscore the broader lesson: additional complexity does not guarantee improved routing outcomes.

5.3 Implications for the Prediction-Application Gap

Beyond explaining individual method performance, our findings contribute to a growing literature questioning the assumption that improved prediction accuracy translates to improved downstream performance. This prediction-application gap has been documented in clinical medicine and recommender systems; we provide evidence that it extends to spatial prediction for routing applications, with specific mechanisms—discontinuous predictions, learned versus encoded spatial structure, and misaligned optimization objectives.

The practical implication is clear: researchers developing spatial prediction methods for routing applications should evaluate on routing outcomes, not prediction accuracy. Our corridor-based ground truth framework provides one approach to such evaluation, measuring actual crime avoidance rather than relying on surrogate metrics. More broadly, the results suggest that method selection should be guided by the specific requirements of the downstream application rather than general-purpose accuracy benchmarks.

The deployment implications extend beyond method selection. Zhang and Bandara (2024) found that 82.3% of pedestrian safety concerns occur during walking, yet current navigation applications do not incorporate safety considerations. Our finding that KDE outperforms ML methods means this unmet demand can be addressed without complex infrastructure—a KDE-based routing system requires only geocoded crime data and a spatial statistics library.

5.4 Spatial Autocorrelation and the Optimality of Explicit Encoding

KDE's advantage has a theoretical foundation in Tobler's First Law of Geography (Tobler, 1970; Miller, 2004): near things are more related than distant things. This manifests as spatial autocorrelation—nearby locations exhibiting similar crime rates—among the most robust findings in environmental criminology (Mohler et al., 2011).

Spatial autocorrelation is quantified through Moran's I statistic (Moran, 1950):

$$I = \frac{N}{W} \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_{i=1}^{N} (x_i - \bar{x})^2} \tag{4}$$

where N is the number of spatial units, w_{ij} are spatial weights (with zeros on the diagonal), W is the sum of all weights, and x_i represents the variable of interest at location i. Values significantly above the expected value under spatial randomness, $E[I] = -1/(N-1)$, indicate positive spatial autocorrelation (clustering), while values significantly below indicate negative autocorrelation (dispersion). Crime data typically exhibit Moran's I values between 0.3 and 0.7, indicating moderate to strong positive spatial autocorrelation (Ratcliffe, 2010). This clustering is theoretically grounded in environmental criminology: crime pattern theory posits that offenses concentrate near offenders' activity nodes and along routine pathways, with crime probability following distance decay from these anchor points (Brantingham & Brantingham, 1993).

KDE directly encodes this structure through its kernel function: the bandwidth parameter specifies the range of spatial dependence, and when this assumed scale approximates the true correlation length in crime data, the resulting risk surfaces respect the underlying spatial structure. Bandwidth selection thus reduces to matching the kernel's spatial scale to the phenomenon's correlation length—a well-studied problem with established solutions (Silverman, 1986).

The mechanisms through which ML methods fail to preserve this structure—discontinuous predictions from tree-based models and misaligned training objectives (see Section 5.1)—can be understood as failures to maintain the spatial autocorrelation that KDE encodes by construction. Graph neural networks partially address the representation challenge through message-passing but face oversmoothing at depth, where node representations converge and local risk variations are obscured (Rusch et al., 2023).

5.5 When Explicit Spatial Encoding Outperforms Machine Learning

Our empirical findings, combined with the theoretical analysis above, suggest conditions under which classical spatial statistics methods may be preferred over machine learning for spatial optimization tasks. We propose the following framework:

Conjecture (Structural Prior Principle): For spatial optimization tasks, explicit encoding of spatial structure is likely to outperform learned representations when:

1. *The spatial autocorrelation structure is well-characterized by established theory (e.g., distance decay in environmental criminology);*
2. *The downstream task optimizes over spatial aggregates (paths, regions) rather than point predictions;*
3. *The optimization objective requires comparing alternatives that differ by small spatial displacements;*
4. *Training data exhibit high spatial sparsity relative to the required output resolution.*

Pedestrian safety routing satisfies all four conditions: crime exhibits theoretically-grounded distance decay (condition 1), routing optimizes over paths (condition 2), alternative routes differ by neighborhood-scale displacements (condition 3), and most street segments experience zero crimes during any training period (condition 4). This explains KDE's substantial advantage over both XGBoost and GNN despite the latter's greater flexibility.

When these conditions are violated—when spatial structure is unknown or non-stationary, when point prediction is the end goal, or when training data are dense—machine learning’s flexibility may prove advantageous.

Before defaulting to machine learning for spatial applications, practitioners should assess whether these conditions are satisfied. If so, classical spatial statistics methods warrant consideration—not as outdated baselines, but as potentially superior approaches whose assumptions align with task requirements.

5.6 Limitations and Future Directions

Several limitations warrant acknowledgment. First, our evaluation is limited to a single city (Chicago), and the performance hierarchy may differ in cities with different crime patterns, street

network structures, or data quality. Replication in additional urban contexts would strengthen confidence in our findings. Second, we focus exclusively on violent crime, which represents a small fraction of total crime incidents; the relative performance of methods may differ for property crime or other offense categories with different spatial distributions. Third, our corridor-based evaluation assumes that crime incidents occurring within R meters of a route pose equivalent risk to the pedestrian, which simplifies the complex relationship between route geometry and actual exposure. More sophisticated exposure models incorporating visibility, escape routes, and temporal dynamics may yield different conclusions.

Fourth, our machine learning implementations do not exhaust the space of possible architectures. Decision-focused learning that optimizes directly for routing outcomes rather than crime count prediction (Elmachtoub & Grigas, 2022) could potentially close the gap with KDE.

Fifth, routing pedestrians away from high-crime neighborhoods—which often correlate with historically marginalized communities—raises equity concerns (O'Donnell, 2019; Huq, 2019). While pedestrian routing differs fundamentally from predictive policing, implementers should consider complementary strategies such as combining safety routing with urban investment initiatives addressing root causes of crime concentration.

Future work should extend this evaluation framework in several directions. Multi-city replication would establish the generalizability of our findings. Incorporation of additional safety factors—street lighting, pedestrian infrastructure, perceived safety from user surveys—would enable more comprehensive safety-aware routing. Real-time routing with dynamic risk updates, incorporating recent incidents and time-of-day effects, represents a natural extension with significant practical value. Finally, user studies evaluating whether pedestrians actually follow recommended routes and whether those routes reduce subjective fear or objective victimization would provide the ultimate validation of safety-aware routing systems.

6. Conclusion

This paper introduced corridor-based ground truth evaluation for pedestrian safety routing and compared five crime risk prediction methods across 36,000 route evaluations per method on Chicago's complete walkable network. The results establish a clear hierarchy: KDE > GNN ≈ XGBoost > Getis-Ord Gi* > Historical Average. KDE achieves 54.5% greater crime avoidance than XGBoost and 53.3% greater than GNN, with a small practical effect (Cohen's d = 0.45) relative to naive shortest-path routing. This hierarchy is consistent across all robustness checks, challenging the assumption that model complexity guarantees superior performance for spatial applications.

The contributions are threefold. First, the corridor-based evaluation framework provides a rigorous, reproducible methodology for assessing routing systems applicable to any city with geocoded crime data and a street network. Second, the empirical comparison—at unprecedented

scale with comprehensive robustness checks—offers actionable guidance: KDE is recommended for deployment due to its superior performance, computational efficiency, and lack of training requirements. Third, our Structural Prior Principle identifies conditions under which explicit spatial encoding outperforms learned representations. We encourage replication across diverse urban contexts; consistent findings would strengthen confidence in KDE's superiority, while divergent results would illuminate boundary conditions for the theoretical framework.

Data and Codes Availability Statement

The code and processed data that support the findings of this study are available from the corresponding author upon reasonable request. The crime incident data are publicly available from the Chicago Data Portal (https://data.cityofchicago.org/, dataset identifier ijzp-q8t2). Street network data were obtained from OpenStreetMap via OSMnx.

Funding

[Funding details removed for double-blind review].

Declaration of Competing Interests

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the author used Claude (Anthropic) for assistance with manuscript formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

CRediT Author Contribution Statement

[Author]: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

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Appendix: Supplementary Materials

This appendix contains supplementary statistical tables and robustness check visualizations. The primary results figures (Figures 1–3) and pairwise comparison table (Table 2) appear in the main text.

Supplementary Tables: Table A.1 (Headline Summary), Table A.2 (Full Descriptive Statistics), Tables A3-A8 (Statistical Tests and Robustness Data)

Supplementary Figures: Figures A1-A7 (Robustness Checks and Sensitivity Analyses), Figure A.8 (Example Route Comparisons)

Table A.1: Headline Summary Statistics

Key performance metrics supporting abstract claims.

Method	Mean Avoided	Win Rate (%)	Mean Detour (%)	KDE Adv. (%)
KDE	0.578	36.0	11.58	—
GNN	0.377	29.7	10.25	53.3
XGBoost	0.374	29.3	8.34	54.5
Getis-Ord Gi*	0.329	27.3	9.32	75.6
Historical Avg	0.045	11.8	1.64	1185.8

Note: N = 36,000 route evaluations per method. KDE Advantage = percentage improvement in mean crimes avoided relative to each alternative method.

Table A.2: Descriptive Statistics by Prediction Method

Method	**N**	**Mean**	**SD**	**Red. (%)**	**Win (%)**	**Loss (%)**	**Detour (%)**
KDE	36,000	0.578	1.282	60.9	36.0	7.3	11.58
GNN	36,000	0.377	1.262	39.7	29.7	13.0	10.25
XGBoost	36,000	0.374	1.253	39.4	29.3	12.5	8.34
Getis-Ord Gi*	36,000	0.329	1.268	34.7	27.3	13.0	9.32
Hist. Avg	36,000	0.045	0.828	4.7	11.8	9.0	1.64

Note: N = 36,000 route evaluations per method (12,000 per month × 3 months). Reduction (%) = percentage of crimes along shortest path that are avoided by the safe route. Win/Loss Rate = percentage of routes where safe route encounters fewer/more corridor crimes than shortest path.

Table A.3: Bootstrap Confidence Intervals

95% CIs via 10,000 bootstrap resamples with bias-corrected acceleration.

Method	**Mean**	**95% CI Lower**	**95% CI Upper**
KDE	0.578	0.564	0.591
GNN	0.377	0.364	0.390
XGBoost	0.374	0.361	0.387
Getis-Ord Gi*	0.329	0.316	0.342
Historical Avg	0.045	0.036	0.053

Note: 10,000 bootstrap resamples. CIs are bias-corrected and accelerated (BCa).

Table A.4: Temporal Consistency: Monthly Breakdown

Mean crimes avoided by month (July-November 2025).

Method	July	October	November	Total	Win Rate (%)
KDE	7,861	7,317	5,613	20,791	36.0
GNN	4,819	5,015	3,727	13,561	29.7
XGBoost	4,690	4,950	3,819	13,459	29.3
Getis-Ord Gi*	4,294	3,700	3,845	11,839	27.3
Historical Avg	696	402	519	1,617	11.8

Note: 12,000 routes evaluated per month. Total crimes avoided across all routes for each month–method combination.

Table A.5: Win Rate Significance Tests

Testing whether each method's win rate significantly exceeds 30%, approximately the expected win rate under random risk assignment given the observed ~45% tie rate. All top-four methods exceed this threshold; no method exceeds 50%.

Method	**N**	**Wins**	**Win (%)**	**p (>10%)**	**Sig**	**p (>30%)**	**Sig**	**p (>50%)**	**Sig**
KDE	36,000	12,977	36.0	< 1e-300	Yes	5.36e-134	Yes	1.000	No
GNN	36,000	10,693	29.7	< 1e-300	Yes	0.892	No	1.000	No
XGBoost	36,000	10,540	29.3	< 1e-300	Yes	0.999	No	1.000	No
Getis-Ord Gi*	36,000	9,837	27.3	< 1e-300	Yes	1.000	No	1.000	No
Hist. Avg	36,000	4,263	11.8	3.84e-30	Yes	1.000	No	1.000	No

Note: One-sided binomial tests. H_0: win rate ≤ threshold. All methods significantly exceed 10% and 30% thresholds; none exceed 50%. Sig (>50%) omitted as all p = 1.0.

Table A.6: Effect Size Analysis (Cohen's d)

Standardized effect sizes. |d| < 0.2 = negligible, 0.2–0.5 = small, 0.5–0.8 = medium, > 0.8 = large.

Method	**Mean**	**SD**	**Cohen's d**	**Interpretation**
KDE	0.578	1.282	0.451	Small
GNN	0.377	1.262	0.299	Small
XGBoost	0.374	1.253	0.298	Small
Getis-Ord Gi*	0.329	1.268	0.259	Small
Historical Avg	0.045	0.828	0.054	Negligible

Note: Cohen's d benchmarks: small = 0.2, medium = 0.5, large = 0.8. Computed as mean / SD (one-sample d against zero).

Table A.7: KDE Relative Performance Improvement

Percentage improvement of KDE over each competing method. Percentages are computed from full-precision mean values before rounding; minor discrepancies may arise when recomputing from the three-decimal-place means reported here.

Comparison	**KDE Total**	**Other Total**	**Improvement (%)**
KDE vs GNN	20,791	13,561	53.3
KDE vs XGBoost	20,791	13,459	54.5
KDE vs Getis-Ord Gi*	20,791	11,839	75.6
KDE vs Historical Avg	20,791	1,617	1185.8

Note: Improvement (%) = (KDE Mean − Other Mean) / Other Mean × 100. N = 36,000 route evaluations per method.

Table A.8: Pareto Frontier Data

Aggregated data for safety-detour tradeoff analysis.

Method	Time Bin	α	Mean Avoided	SD	Detour (%)
Getis-Ord Gi*	Overnight	0	0.000	0.000	0.00
Getis-Ord Gi*	Overnight	25,000	0.229	0.957	0.52
Getis-Ord Gi*	Overnight	50,000	0.263	1.083	1.04
Getis-Ord Gi*	Overnight	75,000	0.304	1.153	1.50
Getis-Ord Gi*	Overnight	100,000	0.331	1.205	2.08
Getis-Ord Gi*	Overnight	150,000	0.353	1.236	3.18
Getis-Ord Gi*	Overnight	200,000	0.387	1.273	4.14
Getis-Ord Gi*	Overnight	250,000	0.387	1.279	4.86
Getis-Ord Gi*	Overnight	350,000	0.390	1.287	6.10
Getis-Ord Gi*	Overnight	500,000	0.414	1.306	7.94
Getis-Ord Gi*	Overnight	750,000	0.426	1.344	9.37
Getis-Ord Gi*	Morning	0	0.000	0.000	0.00
Getis-Ord Gi*	Morning	25,000	0.165	0.899	0.52
Getis-Ord Gi*	Morning	50,000	0.194	0.993	1.04
Getis-Ord Gi*	Morning	75,000	0.237	1.105	1.50
Getis-Ord Gi*	Morning	100,000	0.279	1.145	2.08
Getis-Ord Gi*	Morning	150,000	0.327	1.201	3.18
Getis-Ord Gi*	Morning	200,000	0.357	1.221	4.14
Getis-Ord Gi*	Morning	250,000	0.369	1.232	4.86
Getis-Ord Gi*	Morning	350,000	0.390	1.250	6.10
Getis-Ord Gi*	Morning	500,000	0.403	1.252	7.94
Getis-Ord Gi*	Morning	750,000	0.405	1.260	9.37
Getis-Ord Gi*	Afternoon	0	0.000	0.000	0.00
Getis-Ord Gi*	Afternoon	25,000	0.187	0.737	0.52
Getis-Ord Gi*	Afternoon	50,000	0.224	0.791	1.04
Getis-Ord Gi*	Afternoon	75,000	0.242	0.826	1.50
Getis-Ord Gi*	Afternoon	100,000	0.245	0.855	2.08

Getis-Ord Gi*	Afternoon	150,000	0.260	0.893	3.18
Getis-Ord Gi*	Afternoon	200,000	0.278	0.942	4.14
Getis-Ord Gi*	Afternoon	250,000	0.291	0.960	4.86
Getis-Ord Gi*	Afternoon	350,000	0.322	0.994	6.10
Getis-Ord Gi*	Afternoon	500,000	0.311	1.014	7.94
Getis-Ord Gi*	Afternoon	750,000	0.329	1.026	9.37
Getis-Ord Gi*	Evening	0	0.000	0.000	0.00
Getis-Ord Gi*	Evening	25,000	0.204	0.973	0.52
Getis-Ord Gi*	Evening	50,000	0.218	1.041	1.04
Getis-Ord Gi*	Evening	75,000	0.256	1.087	1.50
Getis-Ord Gi*	Evening	100,000	0.294	1.158	2.08
Getis-Ord Gi*	Evening	150,000	0.340	1.207	3.18
Getis-Ord Gi*	Evening	200,000	0.357	1.248	4.14
Getis-Ord Gi*	Evening	250,000	0.369	1.249	4.86
Getis-Ord Gi*	Evening	350,000	0.385	1.260	6.10
Getis-Ord Gi*	Evening	500,000	0.406	1.300	7.94
Getis-Ord Gi*	Evening	750,000	0.432	1.306	9.37
GNN	Overnight	0	0.000	0.000	0.00
GNN	Overnight	50	0.243	0.975	0.46
GNN	Overnight	100	0.273	1.081	0.97
GNN	Overnight	500	0.367	1.337	4.65
GNN	Overnight	1,000	0.368	1.366	7.92
GNN	Overnight	1,250	0.367	1.361	9.66
GNN	Overnight	1,500	0.358	1.362	10.88
GNN	Overnight	2,500	0.369	1.365	15.94
GNN	Overnight	3,500	0.384	1.371	19.31
GNN	Overnight	5,000	0.386	1.382	23.74
GNN	Overnight	7,500	0.370	1.415	28.65
GNN	Overnight	10,000	0.348	1.447	33.53
GNN	Morning	0	0.000	0.000	0.00

GNN	Morning	50	0.195	0.801	0.42
GNN	Morning	100	0.263	1.027	0.95
GNN	Morning	500	0.361	1.203	4.25
GNN	Morning	1,000	0.365	1.224	7.50
GNN	Morning	1,250	0.370	1.236	8.62
GNN	Morning	1,500	0.379	1.232	9.74
GNN	Morning	2,500	0.373	1.245	12.94
GNN	Morning	3,500	0.367	1.238	15.12
GNN	Morning	5,000	0.364	1.246	18.11
GNN	Morning	7,500	0.363	1.251	23.34
GNN	Morning	10,000	0.362	1.278	26.98
GNN	Afternoon	0	0.000	0.000	0.00
GNN	Afternoon	50	0.174	0.824	0.54
GNN	Afternoon	100	0.200	0.917	1.07
GNN	Afternoon	500	0.294	1.126	5.04
GNN	Afternoon	1,000	0.283	1.151	8.17
GNN	Afternoon	1,250	0.281	1.163	9.45
GNN	Afternoon	1,500	0.274	1.166	10.68
GNN	Afternoon	2,500	0.265	1.196	14.78
GNN	Afternoon	3,500	0.256	1.197	18.20
GNN	Afternoon	5,000	0.264	1.202	21.92
GNN	Afternoon	7,500	0.252	1.205	26.81
GNN	Afternoon	10,000	0.243	1.207	30.92
GNN	Evening	0	0.000	0.000	0.00
GNN	Evening	50	0.257	0.949	0.51
GNN	Evening	100	0.294	1.028	1.02
GNN	Evening	500	0.385	1.166	4.80
GNN	Evening	1,000	0.399	1.200	7.99
GNN	Evening	1,250	0.404	1.204	9.23
GNN	Evening	1,500	0.412	1.211	10.26

GNN	Evening	2,500	0.424	1.237	13.95
GNN	Evening	3,500	0.413	1.222	16.38
GNN	Evening	5,000	0.409	1.225	19.33
GNN	Evening	7,500	0.398	1.249	23.26
GNN	Evening	10,000	0.416	1.263	26.01
Hist. Avg	Overnight	0	0.000	0.000	0.00
Hist. Avg	Overnight	25,000	-0.006	0.727	0.77
Hist. Avg	Overnight	50,000	-0.012	0.785	1.05
Hist. Avg	Overnight	75,000	0.010	0.790	1.23
Hist. Avg	Overnight	100,000	0.006	0.794	1.33
Hist. Avg	Overnight	150,000	0.009	0.818	1.44
Hist. Avg	Overnight	200,000	0.008	0.814	1.56
Hist. Avg	Overnight	250,000	0.009	0.813	1.67
Hist. Avg	Overnight	350,000	-0.002	0.850	1.75
Hist. Avg	Overnight	500,000	-0.005	0.851	1.98
Hist. Avg	Overnight	750,000	-0.005	0.851	2.08
Hist. Avg	Morning	0	0.000	0.000	0.00
Hist. Avg	Morning	25,000	0.047	0.666	0.77
Hist. Avg	Morning	50,000	0.060	0.716	1.05
Hist. Avg	Morning	75,000	0.076	0.727	1.23
Hist. Avg	Morning	100,000	0.068	0.721	1.33
Hist. Avg	Morning	150,000	0.070	0.732	1.44
Hist. Avg	Morning	200,000	0.069	0.737	1.56
Hist. Avg	Morning	250,000	0.069	0.737	1.67
Hist. Avg	Morning	350,000	0.068	0.738	1.75
Hist. Avg	Morning	500,000	0.071	0.739	1.98
Hist. Avg	Morning	750,000	0.071	0.739	2.08
Hist. Avg	Afternoon	0	0.000	0.000	0.00
Hist. Avg	Afternoon	25,000	0.043	0.635	0.77
Hist. Avg	Afternoon	50,000	0.047	0.678	1.05

Hist. Avg	Afternoon	75,000	0.043	0.682	1.23
Hist. Avg	Afternoon	100,000	0.044	0.696	1.33
Hist. Avg	Afternoon	150,000	0.046	0.700	1.44
Hist. Avg	Afternoon	200,000	0.050	0.687	1.56
Hist. Avg	Afternoon	250,000	0.051	0.688	1.67
Hist. Avg	Afternoon	350,000	0.049	0.691	1.75
Hist. Avg	Afternoon	500,000	0.047	0.695	1.98
Hist. Avg	Afternoon	750,000	0.047	0.695	2.08
Hist. Avg	Evening	0	0.000	0.000	0.00
Hist. Avg	Evening	25,000	0.035	0.791	0.77
Hist. Avg	Evening	50,000	0.032	0.835	1.05
Hist. Avg	Evening	75,000	0.037	0.848	1.23
Hist. Avg	Evening	100,000	0.036	0.868	1.33
Hist. Avg	Evening	150,000	0.035	0.874	1.44
Hist. Avg	Evening	200,000	0.039	0.868	1.56
Hist. Avg	Evening	250,000	0.037	0.864	1.67
Hist. Avg	Evening	350,000	0.037	0.864	1.75
Hist. Avg	Evening	500,000	0.036	0.866	1.98
Hist. Avg	Evening	750,000	0.036	0.866	2.08
Hotspot	Overnight	0	0.000	0.000	0.00
Hotspot	Overnight	25,000	0.294	1.107	1.90
Hotspot	Overnight	50,000	0.377	1.290	4.32
Hotspot	Overnight	75,000	0.380	1.322	6.20
Hotspot	Overnight	100,000	0.387	1.361	7.78
Hotspot	Overnight	150,000	0.389	1.391	10.37
Hotspot	Overnight	200,000	0.381	1.406	12.18
Hotspot	Overnight	250,000	0.382	1.430	13.25
Hotspot	Overnight	350,000	0.382	1.426	16.08
Hotspot	Overnight	500,000	0.390	1.437	21.09
Hotspot	Overnight	750,000	0.389	1.432	27.02

Hotspot	Morning	0	0.000	0.000	0.00
Hotspot	Morning	25,000	0.290	1.069	1.90
Hotspot	Morning	50,000	0.350	1.182	4.32
Hotspot	Morning	75,000	0.363	1.210	6.20
Hotspot	Morning	100,000	0.376	1.228	7.78
Hotspot	Morning	150,000	0.366	1.241	10.37
Hotspot	Morning	200,000	0.370	1.248	12.18
Hotspot	Morning	250,000	0.372	1.254	13.25
Hotspot	Morning	350,000	0.364	1.254	16.08
Hotspot	Morning	500,000	0.339	1.224	21.09
Hotspot	Morning	750,000	0.364	1.242	27.02
Hotspot	Afternoon	0	0.000	0.000	0.00
Hotspot	Afternoon	25,000	0.261	0.921	1.90
Hotspot	Afternoon	50,000	0.288	0.996	4.32
Hotspot	Afternoon	75,000	0.300	1.057	6.20
Hotspot	Afternoon	100,000	0.313	1.071	7.78
Hotspot	Afternoon	150,000	0.332	1.103	10.37
Hotspot	Afternoon	200,000	0.329	1.118	12.18
Hotspot	Afternoon	250,000	0.323	1.130	13.25
Hotspot	Afternoon	350,000	0.312	1.128	16.08
Hotspot	Afternoon	500,000	0.305	1.129	21.09
Hotspot	Afternoon	750,000	0.301	1.159	27.02
Hotspot	Evening	0	0.000	0.000	0.00
Hotspot	Evening	25,000	0.320	1.080	1.90
Hotspot	Evening	50,000	0.394	1.188	4.32
Hotspot	Evening	75,000	0.424	1.228	6.20
Hotspot	Evening	100,000	0.415	1.267	7.78
Hotspot	Evening	150,000	0.433	1.272	10.37
Hotspot	Evening	200,000	0.425	1.286	12.18
Hotspot	Evening	250,000	0.421	1.292	13.25

Hotspot	Evening	350,000	0.413	1.279	16.08
Hotspot	Evening	500,000	0.409	1.309	21.09
Hotspot	Evening	750,000	0.421	1.323	27.02
KDE	Overnight	0	0.000	0.000	0.00
KDE	Overnight	25,000	0.379	1.129	1.31
KDE	Overnight	50,000	0.446	1.200	2.48
KDE	Overnight	75,000	0.471	1.238	3.76
KDE	Overnight	100,000	0.468	1.239	4.75
KDE	Overnight	150,000	0.482	1.272	6.54
KDE	Overnight	200,000	0.509	1.333	7.96
KDE	Overnight	250,000	0.509	1.339	9.42
KDE	Overnight	350,000	0.500	1.341	11.75
KDE	Overnight	500,000	0.523	1.380	15.69
KDE	Overnight	750,000	0.532	1.405	20.56
KDE	Morning	0	0.000	0.000	0.00
KDE	Morning	25,000	0.392	1.121	1.31
KDE	Morning	50,000	0.427	1.173	2.48
KDE	Morning	75,000	0.461	1.197	3.76
KDE	Morning	100,000	0.473	1.214	4.75
KDE	Morning	150,000	0.475	1.195	6.54
KDE	Morning	200,000	0.500	1.196	7.96
KDE	Morning	250,000	0.511	1.200	9.42
KDE	Morning	350,000	0.526	1.187	11.75
KDE	Morning	500,000	0.549	1.187	15.69
KDE	Morning	750,000	0.566	1.200	20.56
KDE	Afternoon	0	0.000	0.000	0.00
KDE	Afternoon	25,000	0.358	0.943	1.31
KDE	Afternoon	50,000	0.408	0.991	2.48
KDE	Afternoon	75,000	0.435	1.006	3.76
KDE	Afternoon	100,000	0.462	1.040	4.75

KDE	Afternoon	150,000	0.482	1.060	6.54
KDE	Afternoon	200,000	0.493	1.057	7.96
KDE	Afternoon	250,000	0.510	1.052	9.42
KDE	Afternoon	350,000	0.519	1.045	11.75
KDE	Afternoon	500,000	0.536	1.056	15.69
KDE	Afternoon	750,000	0.555	1.054	20.56
KDE	Evening	0	0.000	0.000	0.00
KDE	Evening	25,000	0.419	1.087	1.31
KDE	Evening	50,000	0.496	1.162	2.48
KDE	Evening	75,000	0.520	1.202	3.76
KDE	Evening	100,000	0.543	1.202	4.75
KDE	Evening	150,000	0.589	1.218	6.54
KDE	Evening	200,000	0.627	1.226	7.96
KDE	Evening	250,000	0.634	1.236	9.42
KDE	Evening	350,000	0.658	1.268	11.75
KDE	Evening	500,000	0.688	1.256	15.69
KDE	Evening	750,000	0.713	1.267	20.56
XGBoost	Overnight	0	0.000	0.000	0.00
XGBoost	Overnight	50	0.238	0.989	0.47
XGBoost	Overnight	100	0.279	1.096	0.99
XGBoost	Overnight	500	0.392	1.357	4.97
XGBoost	Overnight	1,000	0.380	1.376	8.65
XGBoost	Overnight	1,250	0.381	1.386	10.13
XGBoost	Overnight	1,500	0.367	1.362	11.73
XGBoost	Overnight	2,500	0.371	1.363	16.85
XGBoost	Overnight	3,500	0.378	1.362	20.66
XGBoost	Overnight	5,000	0.379	1.366	24.56
XGBoost	Overnight	7,500	0.363	1.420	30.08
XGBoost	Overnight	10,000	0.344	1.438	34.36
XGBoost	Overnight	15,000	0.365	1.467	41.46

XGBoost	Morning	0	0.000	0.000	0.00
XGBoost	Morning	50	0.198	0.810	0.42
XGBoost	Morning	100	0.278	1.034	0.96
XGBoost	Morning	500	0.370	1.212	4.46
XGBoost	Morning	1,000	0.383	1.234	7.82
XGBoost	Morning	1,250	0.382	1.239	9.11
XGBoost	Morning	1,500	0.390	1.231	10.38
XGBoost	Morning	2,500	0.369	1.241	13.53
XGBoost	Morning	3,500	0.373	1.241	15.68
XGBoost	Morning	5,000	0.377	1.253	19.00
XGBoost	Morning	7,500	0.367	1.255	24.10
XGBoost	Morning	10,000	0.364	1.281	27.20
XGBoost	Morning	15,000	0.372	1.289	33.60
XGBoost	Afternoon	0	0.000	0.000	0.00
XGBoost	Afternoon	50	0.172	0.834	0.58
XGBoost	Afternoon	100	0.200	0.910	1.16
XGBoost	Afternoon	500	0.301	1.087	5.35
XGBoost	Afternoon	1,000	0.306	1.131	9.07
XGBoost	Afternoon	1,250	0.292	1.145	10.64
XGBoost	Afternoon	1,500	0.277	1.150	12.26
XGBoost	Afternoon	2,500	0.263	1.173	16.94
XGBoost	Afternoon	3,500	0.263	1.173	20.66
XGBoost	Afternoon	5,000	0.267	1.187	24.66
XGBoost	Afternoon	7,500	0.226	1.191	30.15
XGBoost	Afternoon	10,000	0.227	1.211	34.69
XGBoost	Afternoon	15,000	0.225	1.251	40.75
XGBoost	Evening	0	0.000	0.000	0.00
XGBoost	Evening	50	0.253	0.960	0.55
XGBoost	Evening	100	0.314	1.059	1.09
XGBoost	Evening	500	0.407	1.174	5.44

XGBoost	Evening	1,000	0.432	1.230	9.01
XGBoost	Evening	1,250	0.440	1.229	10.13
XGBoost	Evening	1,500	0.430	1.227	11.43
XGBoost	Evening	2,500	0.412	1.251	14.94
XGBoost	Evening	3,500	0.422	1.230	17.80
XGBoost	Evening	5,000	0.408	1.241	21.04
XGBoost	Evening	7,500	0.413	1.259	24.65
XGBoost	Evening	10,000	0.422	1.275	27.98
XGBoost	Evening	15,000	0.414	1.315	32.79

Note: Selected operating points shown ($\alpha \in \{0, 50, 100, 250, 500, 750, 1000, 2500, 5000, 10000\}$). Full dataset (110 rows) available in supplementary material. Time bins: Overnight = 00:00–05:59, Morning = 06:00–11:59, Afternoon = 12:00–17:59, Evening = 18:00–23:59.

Table A.9: Mean Crimes Avoided by Community Area and Prediction Method

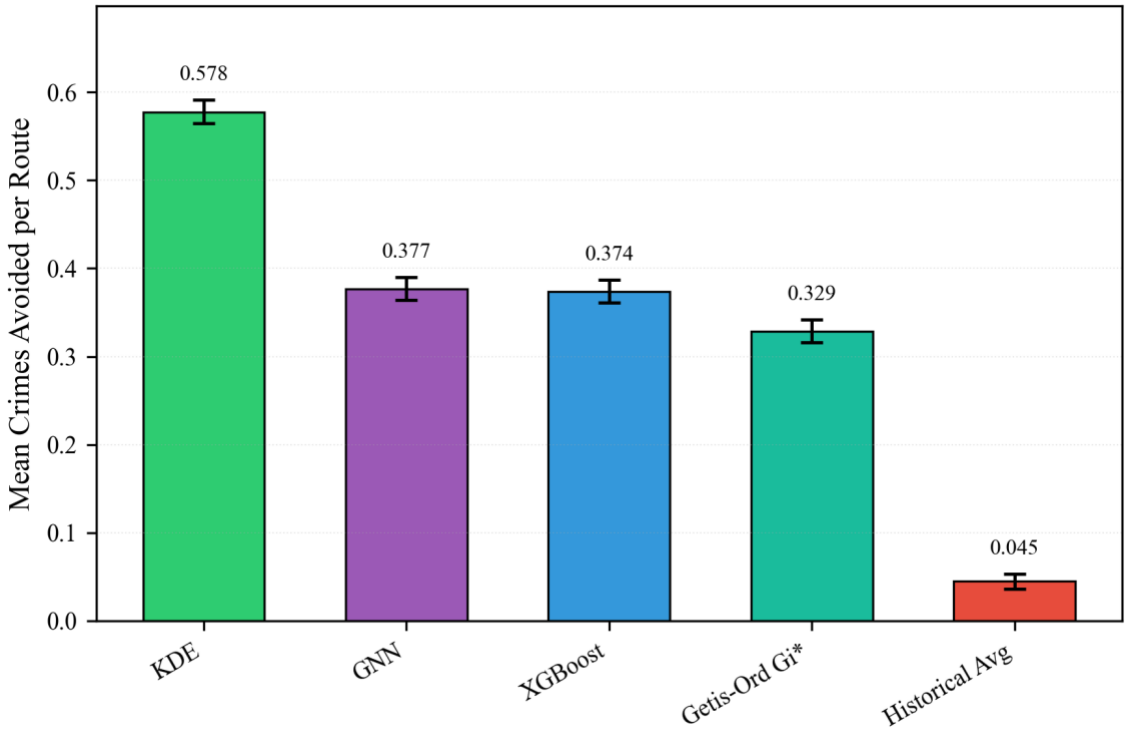
Community	N	KDE	GNN	XGBoost	Getis-Ord Gi*	Hist. Avg
Rogers Park	108	0.333	0.287	0.269	0.194	-0.009
West Ridge	324	0.176	0.046	0.074	0.096	0.019
Uptown	108	0.287	0.352	0.287	0.343	0.185
Lincoln Square	288	0.149	0.076	0.056	0.101	0.003
North Center	288	0.392	0.260	0.240	0.299	0.097
Lake View	288	0.757	0.625	0.639	0.719	0.306
Lincoln Park	288	0.253	0.208	0.219	0.135	0.104
Near North Side	2,736	0.980	0.813	0.799	0.755	0.120
Norwood Park	180	0.072	-0.011	-0.006	0.000	0.011
Jefferson Park	252	0.020	0.020	0.032	0.032	-0.008
Forest Glen	144	0.049	0.014	0.035	0.035	0.049
North Park	108	0.167	-0.009	0.037	-0.009	0.009
Albany Park	432	0.296	0.120	0.123	0.192	0.065
Portage Park	324	0.022	0.034	0.046	-0.052	-0.052
Irving Park	468	0.256	0.143	0.154	0.162	0.075
Dunning	72	0.083	0.083	0.083	0.000	0.042
Montclare	108	0.130	0.028	-0.009	-0.102	0.009
Belmont Cragin	396	0.285	0.033	0.063	0.174	-0.056
Hermosa	108	0.435	0.315	0.204	0.148	0.000
Avondale	216	-0.009	-0.065	-0.097	-0.014	0.000
Logan Square	288	0.444	0.208	0.194	0.170	0.066
Humboldt Park	360	0.633	0.422	0.436	0.439	0.228
West Town	1,116	0.464	0.272	0.282	0.209	0.248
Austin	1,548	0.483	0.335	0.313	0.124	0.099
West Garfield Park	72	0.597	0.458	0.417	0.444	0.153
East Garfield Park	288	0.306	0.222	0.243	-0.257	-0.045
Near West Side	1,908	0.577	0.311	0.303	0.366	-0.098
North Lawndale	144	0.583	0.340	0.347	0.354	0.090

South Lawndale	216	0.546	0.259	0.384	0.264	0.046
Lower West Side	252	0.302	0.083	0.067	0.095	-0.048
Loop	2,196	1.257	1.056	1.055	1.052	0.092
Near South Side	72	0.542	0.292	0.278	0.361	0.000
Armour Square	108	0.231	0.139	0.148	0.157	0.009
Douglas	252	0.619	0.444	0.444	0.397	-0.298
Fuller Park	72	0.889	0.514	0.486	0.625	0.486
Grand Boulevard	180	0.272	0.244	0.244	-0.033	0.056
Kenwood	72	0.514	0.389	0.347	0.194	0.278
Washington Park	1,476	0.743	0.531	0.495	0.488	0.067
Hyde Park	108	0.296	0.231	0.231	0.287	0.167
Woodlawn	288	1.222	0.986	0.969	0.979	-0.087
South Shore	288	0.316	0.198	0.194	0.062	0.014
Chatham	324	0.448	0.426	0.414	0.340	0.170
Avalon Park	36	0.000	-0.194	-0.194	0.000	0.000
South Chicago	72	0.986	0.597	0.667	-0.014	-0.250
Burnside	72	0.319	0.306	0.264	0.222	0.167
Calumet Heights	180	0.117	0.106	0.067	0.006	0.022
Roseland	576	0.387	0.306	0.300	0.245	0.014
Pullman	36	-0.028	0.000	0.000	0.028	0.000
South Deering	72	0.014	-0.056	-0.042	0.014	-0.069
East Side	144	0.083	0.069	0.062	0.028	0.028
West Pullman	288	0.556	0.358	0.378	0.299	0.007
Riverdale	36	-0.028	-0.306	-0.250	0.083	-0.250
Hegewisch	36	0.000	0.000	-0.083	0.000	0.000
Garfield Ridge	288	0.215	0.087	0.076	0.028	0.003
Archer Heights	252	0.218	0.107	0.075	0.040	0.008
Brighton Park	288	0.323	0.233	0.198	0.045	0.014
Mckinley Park	108	0.046	0.074	-0.046	0.065	0.046
Bridgeport	252	0.532	0.417	0.425	0.028	0.234

New City	576	0.330	0.146	0.160	0.118	-0.045
Gage Park	288	0.500	0.267	0.309	0.229	0.115
Clearing	36	0.167	-0.250	-0.250	-0.028	0.000
West Lawn	252	0.099	0.032	0.040	0.083	-0.004
Chicago Lawn	432	0.345	0.181	0.238	0.137	0.016
West Englewood	468	0.833	0.675	0.688	0.611	0.096
Englewood	7,056	0.527	0.188	0.189	0.163	0.035
Greater Grand Crossing	3,132	0.953	0.711	0.699	0.546	-0.018
Ashburn	504	0.222	0.135	0.121	0.046	-0.002
Auburn Gresham	432	0.623	0.417	0.414	0.417	0.093
Beverly	432	0.088	0.021	0.037	0.032	0.009
Washington Heights	216	0.204	0.023	-0.046	0.065	0.051
Mount Greenwood	36	0.000	-0.056	0.000	0.000	0.000
Morgan Park	288	0.052	-0.003	0.007	-0.035	0.007
Ohare	36	0.167	-0.028	-0.028	0.139	0.167
Edgewater	216	0.634	0.347	0.505	0.597	0.023

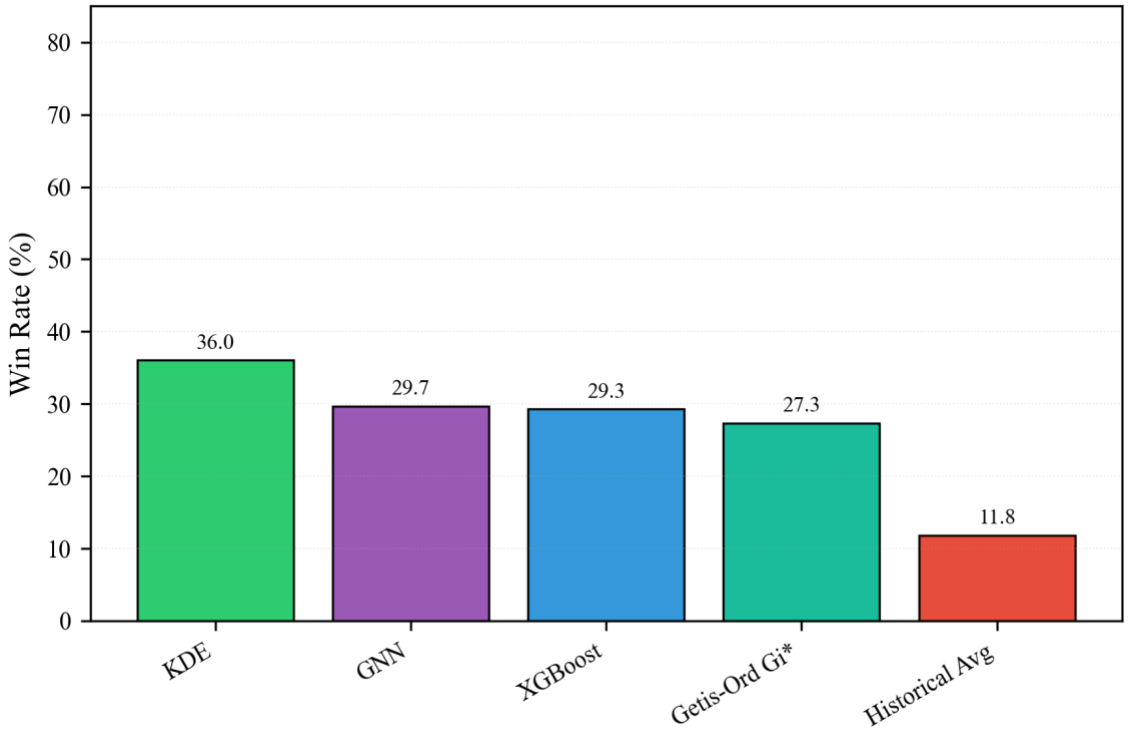
Note: Values represent mean crimes avoided per route for each prediction method across all evaluation conditions within each community area. N = number of route evaluations per method. Communities sorted by area number. Only community areas with at least one route evaluation are included.

Figure 1: Mean Crimes Avoided by Prediction Method



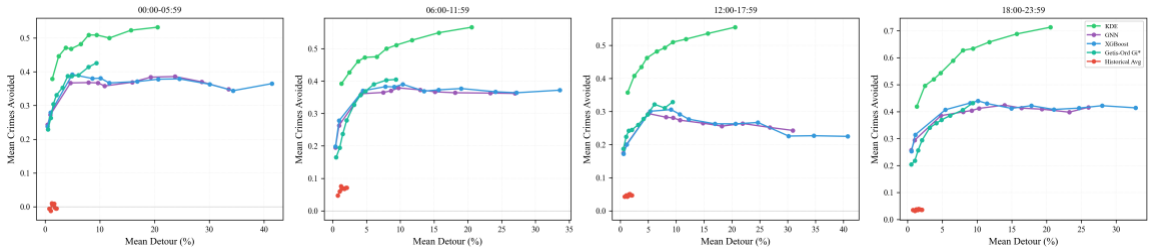
Mean crimes avoided per route by prediction method with 95% confidence intervals. KDE achieves highest mean (0.578) with clear separation from all other methods.

Figure 2: Win Rate by Prediction Method



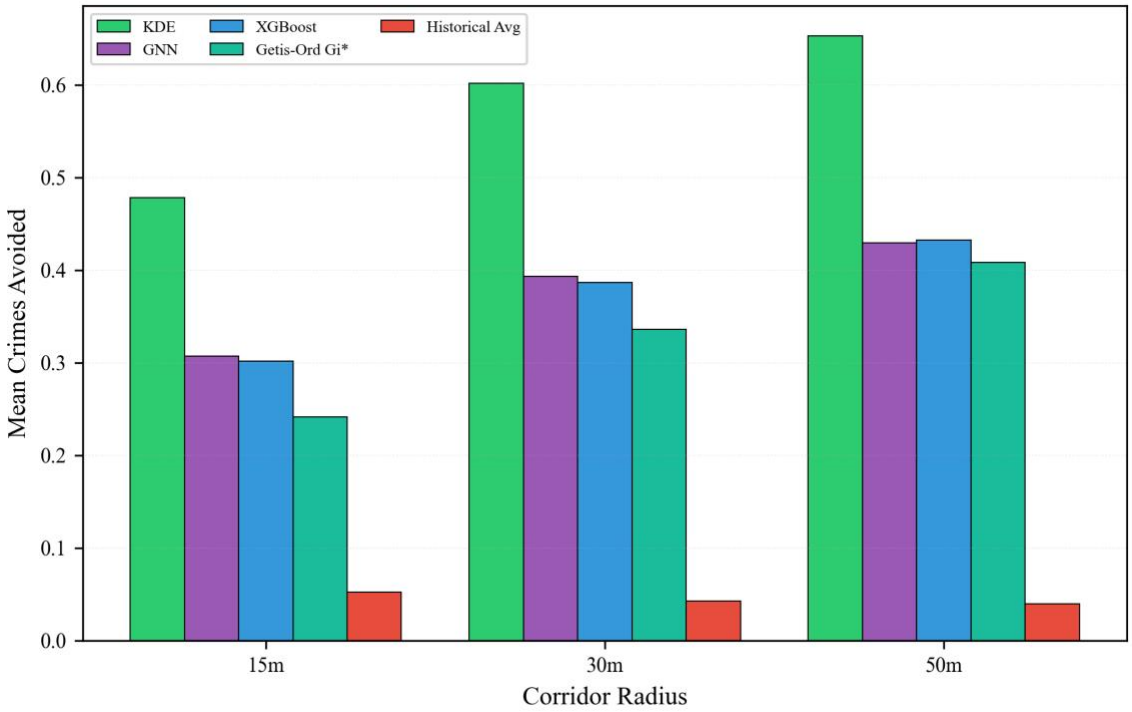
Proportion where safety-aware route avoided more crimes. KDE achieves 36.0% win rate, leading all methods.

Figure 3: Pareto Frontier



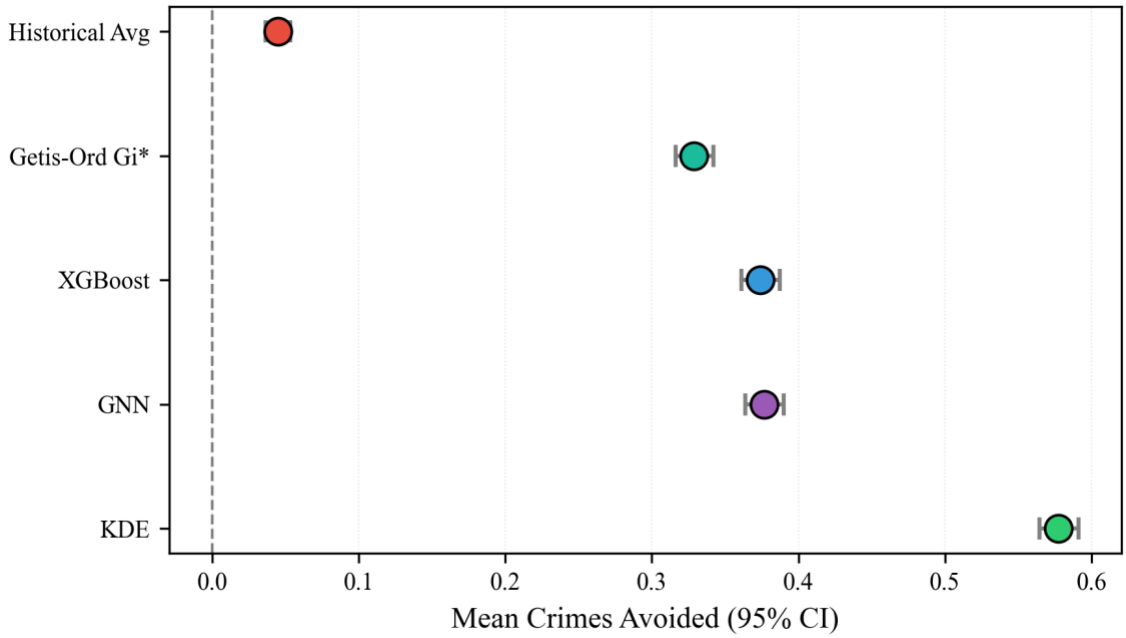
Mean crimes avoided per route vs. detour percentage. KDE dominates at all operating points.

Figure A.1: Robustness Across Corridor Radii



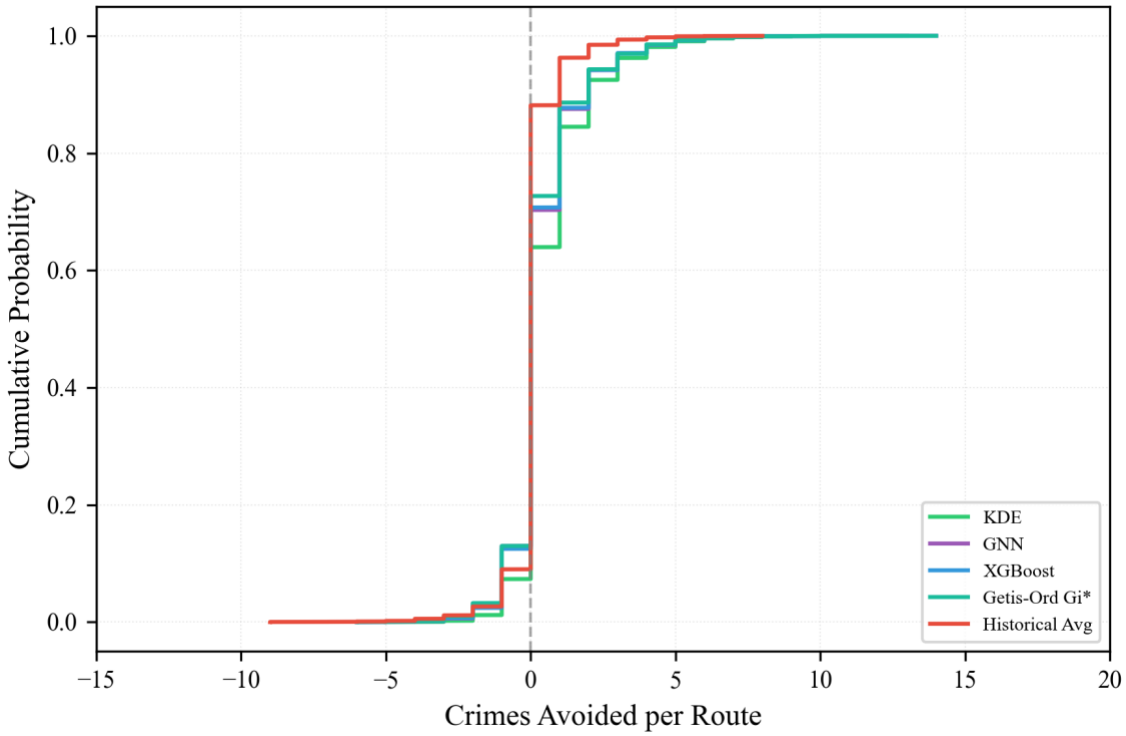
Mean crimes avoided at 15m, 30m, 50m radii. Method ranking consistent.

Figure A.2: Bootstrap 95% Confidence Intervals



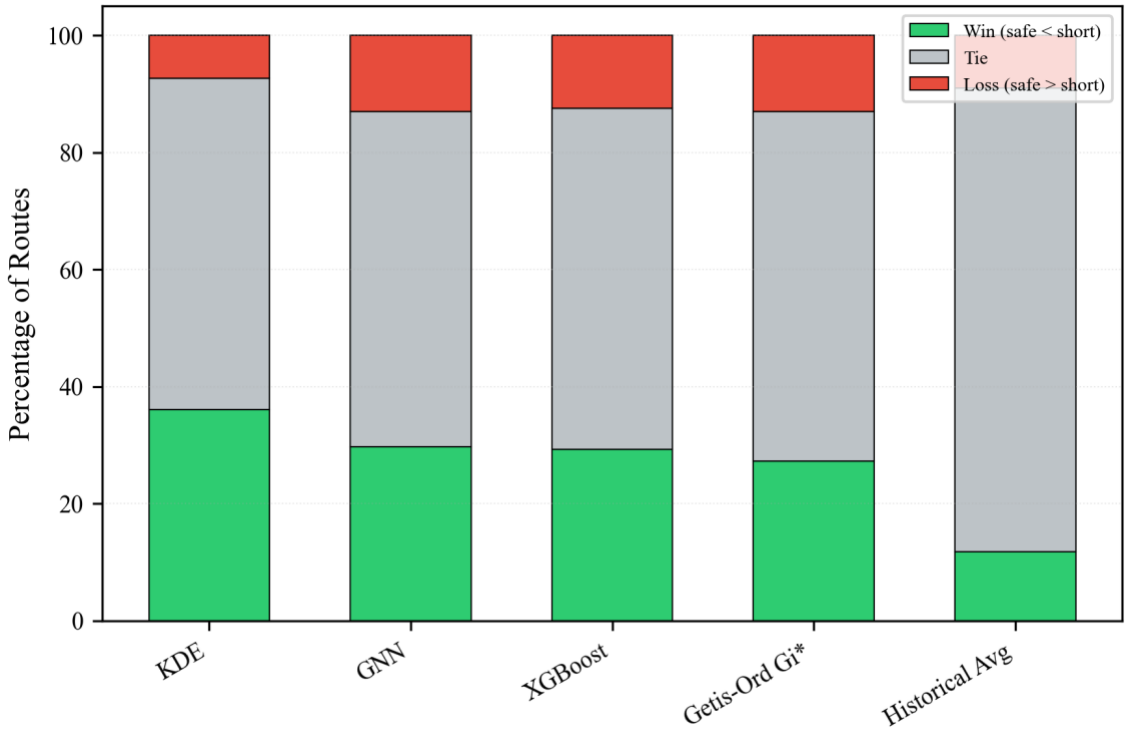
Forest plot. KDE CI does not overlap with any other method.

Figure A.3: Cumulative Distribution Functions



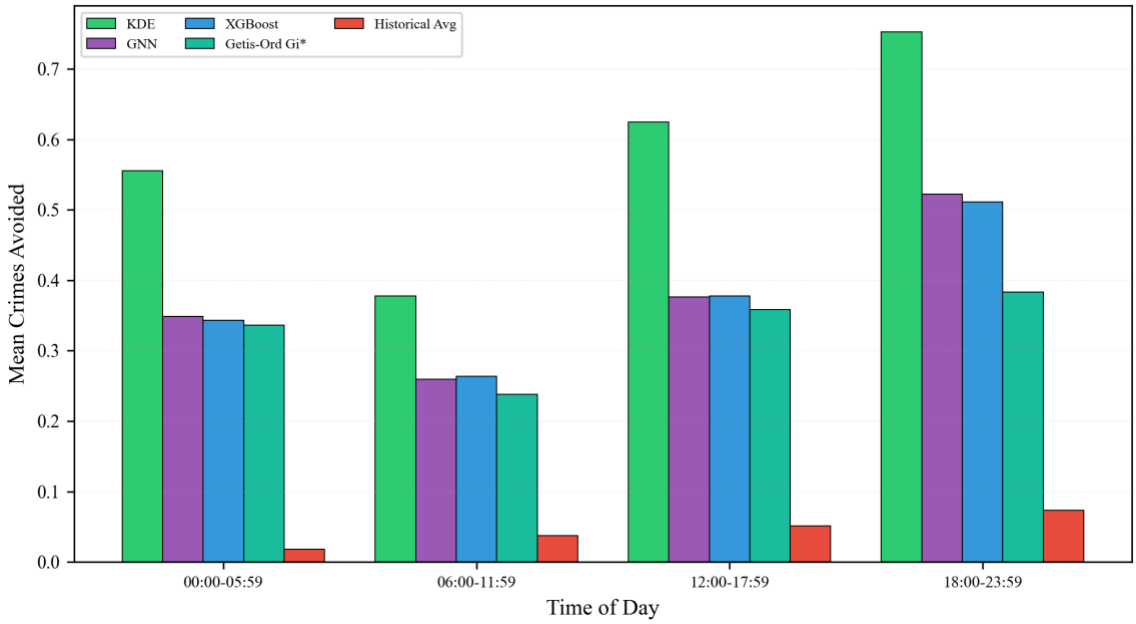
Empirical CDFs showing KDE stochastic dominance.

Figure A.4: Win/Tie/Loss Breakdown



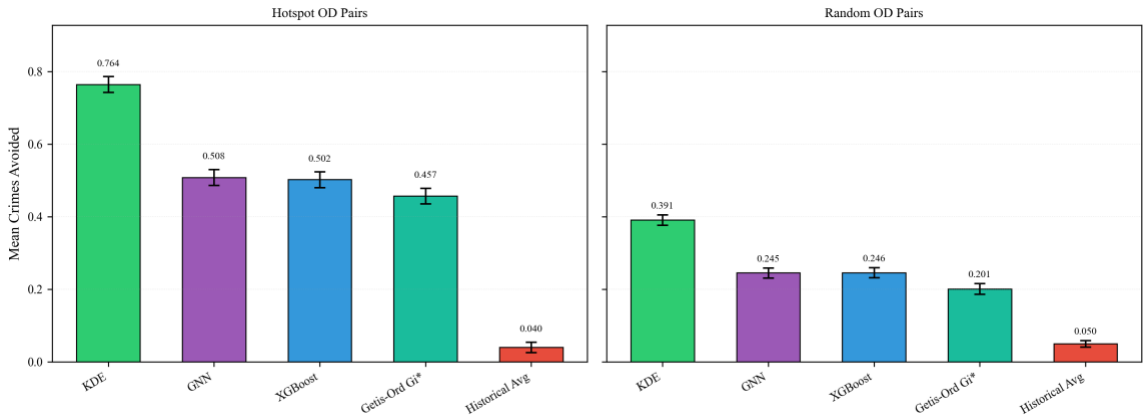
KDE achieves lowest loss rate among all methods.

Figure A.5: Performance Across Time-of-Day Bins



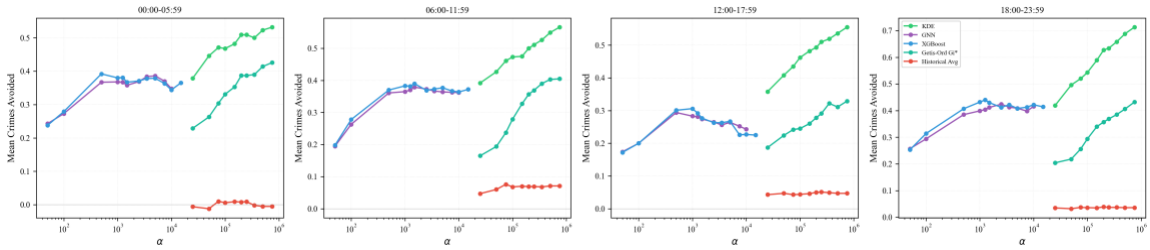
Crime avoidance varies across time-of-day bins. Method ranking consistent across all four bins.

Figure A.6: Hotspot vs. Random OD Pairs



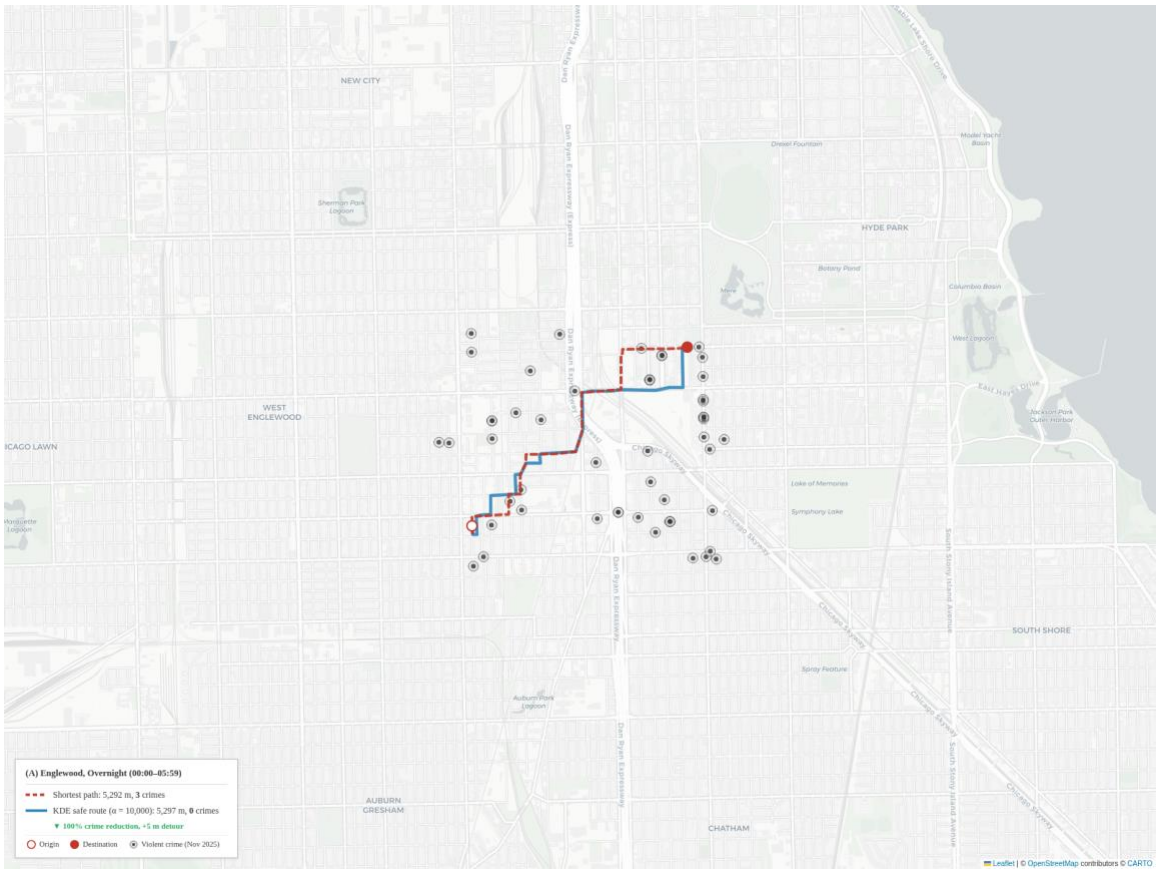
Hotspot OD pairs show higher crime avoidance; random OD pairs traverse low-crime areas with less differentiation.

Figure A.7: Alpha Parameter Sensitivity

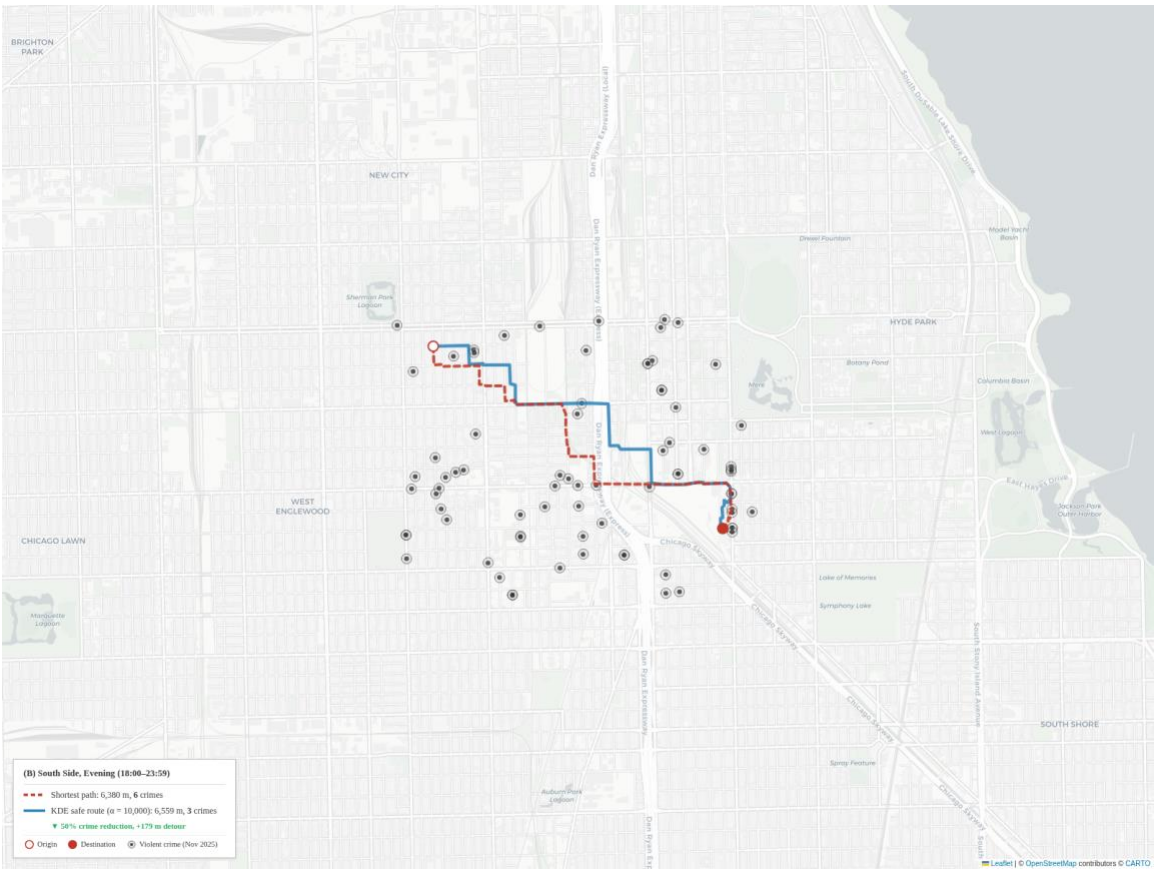


Performance vs. risk aversion parameter. Diminishing returns at high alpha.

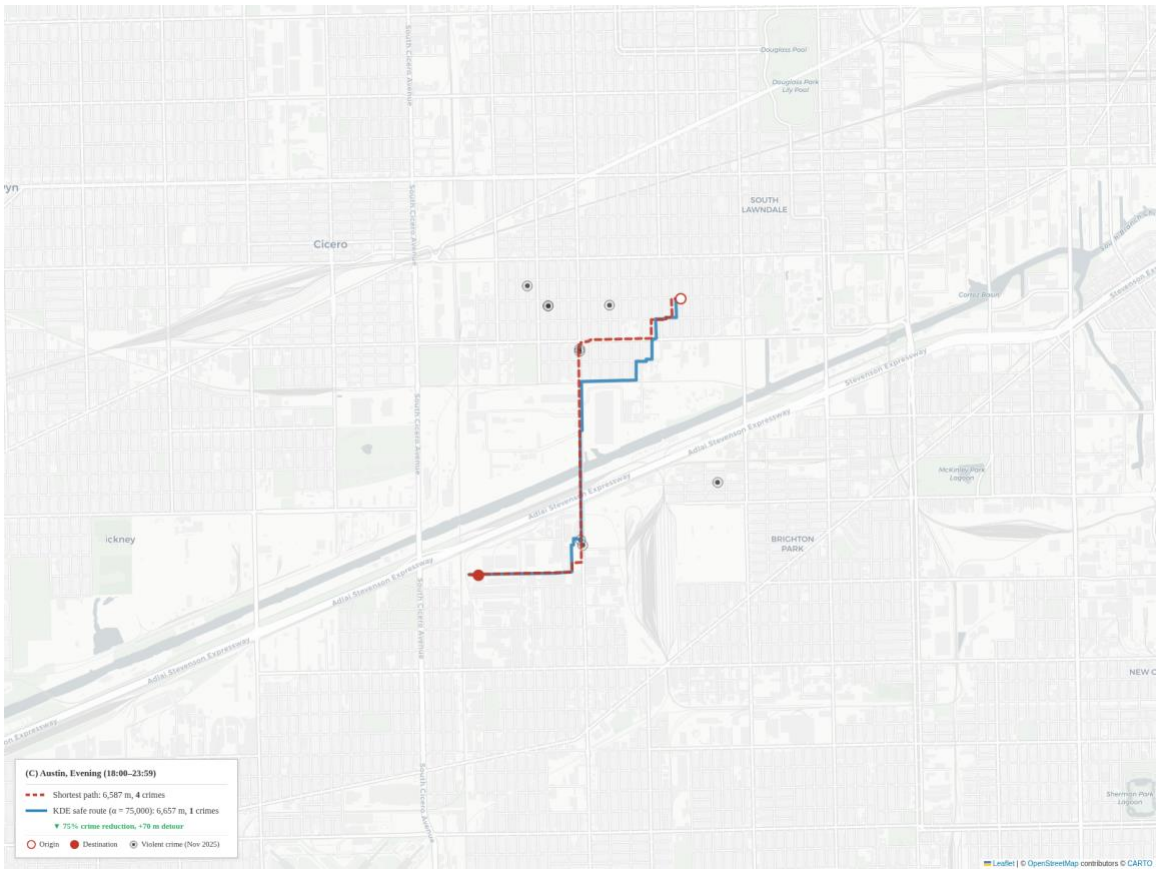
Figure A.8: Spatial Analysis of Crime Avoidance by Community Area



(A) Englewood, Overnight (00:00–05:59). Shortest path: 5,292 m, 3 corridor crimes. KDE safe route (α = 10,000): 5,297 m, 0 crimes. 100% crime reduction, +5 m detour.



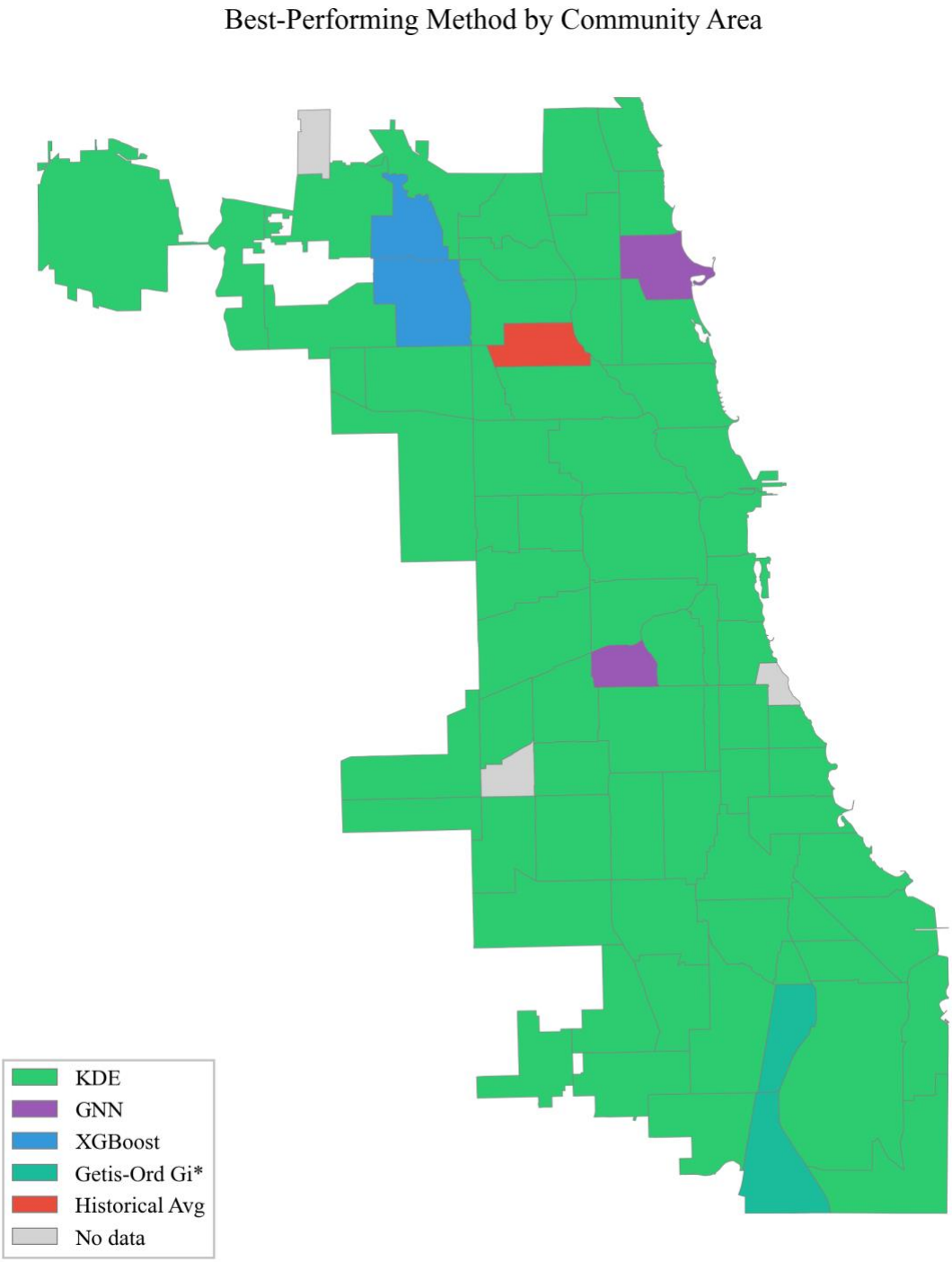
(B) South Side, Evening (18:00–23:59). Shortest path: 6,380 m, 6 corridor crimes. KDE safe route (α = 10,000): 6,559 m, 3 crimes. 50% crime reduction, +179 m detour.



(C) Austin, Evening (18:00–23:59). Shortest path: 6,587 m, 4 corridor crimes. KDE safe route (α = 75,000): 6,657 m, 1 crime. 75% crime reduction, +70 m detour.

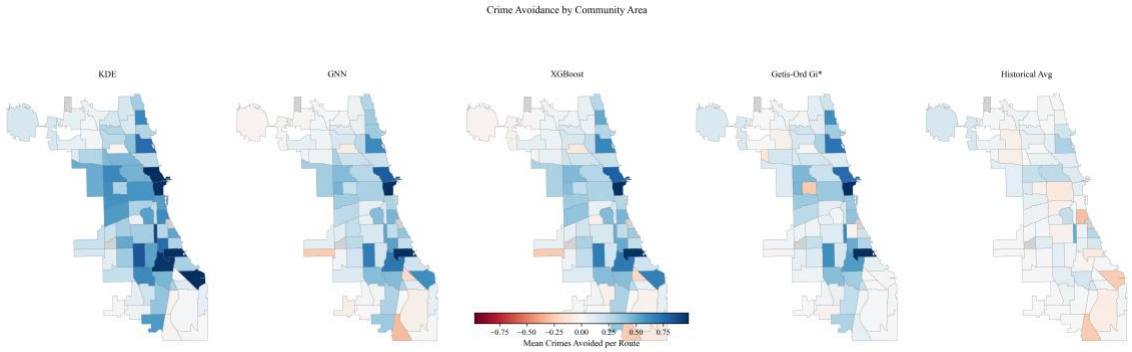
Dashed red: shortest path; solid blue: KDE safe route. Grey dots indicate violent crime incidents during the November 2025 evaluation period (R = 50 m corridor). Routes generated using Dijkstra's algorithm with cost(e) = length(e) + α · log(1 + r̂(e)). Note: map lines delineate study areas and do not necessarily depict accepted national boundaries.

Figure A.9: Best-Performing Method by Community Area



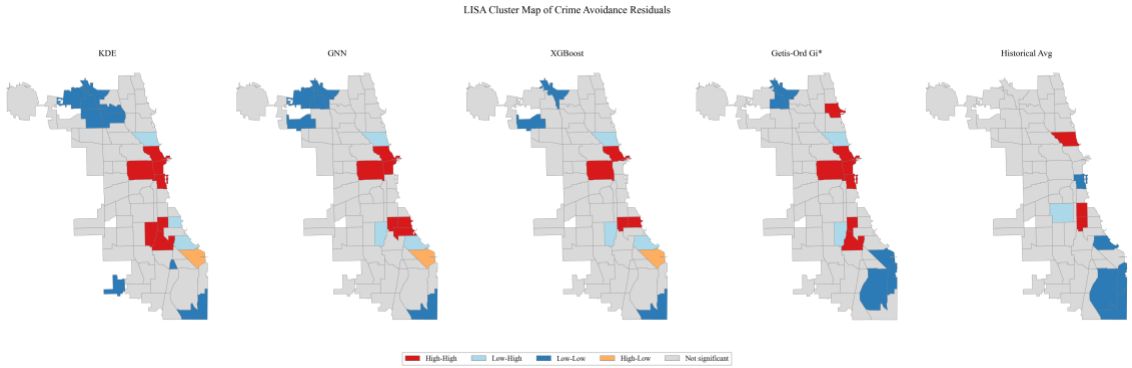
Choropleth map showing the best-performing prediction method per Chicago community area. KDE dominates in the majority of areas.

Figure A.10: Crime Avoidance by Community Area and Prediction Method



Five-panel choropleth maps showing mean crimes avoided per route across Chicago community areas for each prediction method. Darker shading indicates higher crime avoidance. KDE shows the broadest spatial coverage of high crime avoidance.

Figure A.11: LISA Cluster Map of Crime Avoidance Residuals



Local Moran's I cluster analysis of crime avoidance residuals by prediction method. Red (High-High) and blue (Low-Low) indicate statistically significant spatial clusters; orange (High-Low) and light blue (Low-High) indicate spatial outliers. Grey areas are not significant.